Quantum Theory and Locality

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QUANTUM THEORY AND LOCALITY

PHILOSOPHY, PHYSICS, AND THE INTERCONNECTED UNIVERSE



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Preface

This book is being written as an online work in progress. It examines the relationship between quantum theory and the principle of locality — the idea that objects are only directly influenced by their immediate surroundings. Beginning with the foundational formalism of quantum mechanics, the book does not trace the historical and conceptual development of the Einstein–Podolsky–Rosen (EPR) argument (Einstein et al. (1935)) but considers a formal restatement of quantum theory as a generalised probability theory and then Bell's theorem (Bell (1964)) as a statement about empirical locality.

The experimental tests of Bell inequalities that followed will be examined only briefly.

While briefly surveying a range of interpretive positions of quantum theory itself, the book focusses on a formulation that is explicitly possibilist. This theory has been developed by Ruth Kastner (Kastner (2022)) but here it is recognised as a theory that includes physics that is not part of standard quantum theory. The reformulation presented here will emphasise possibility, probability and events. This then informs the discussion on nonlocality, realism, and what quantum theory implies about the structure of physical reality.

The book deviates from most presentations in giving weight to an objective probabilistic interpretation of quantum theory, inspired by Kochen (2015) and to the Relativistic Transactional Interpretation of quantum theory (Kastner (2022)). The book will argue that the Relativistic Transactional Interpretation provides the best available account of events and the nonlocal correlations predicted by quantum

theory and observed in experiments. The book will also explore the implications of these ideas for our understanding of space, time, and causality.

Ontologically, the main innovation is the enlargement of the notion of reality to include not only actual events but also the potentialities described by the quantum formalism that underpins this actualisation. These potentialities are argued to be not merely epistemic but are things in the Real Sphere, and they can influence actual events. This challenges classical intuitions about the nature of reality and causation, suggesting that the universe is fundamentally interconnected in terms of possibilities that give rise to empirical spacetime.

These ontological implications are taken up in a companion book Existence and Being.

Part I.

Foundations

1. A Probabilistic Reconstruction of Quantum Mechanics

In this chapter familiarity with standard non-relativistic quantum mechanics (Bohm (1951)) will be assumed. This will provide the background for the reconstruction of the theory that is a generalisation of classical probability of events.

The mathematical formulation to be discussed here is a modification of Kochen's Reconstruction of Quantum Mechanics (Kochen 2015). It concentrates on a mathematical formulation using a union of σ-algebras that Kochen calls a **complex**. In Kochen's reformulation, it is claimed through the interaction of a quantum system under consideration with another system that properties gain physical values. However this interaction has the same status and faults as the measurement process in standard quantum mechanics. Such properties are said by Kochen to be **relational** or **extrinsic**, as opposed to fixed intrinsic properties such as the charge of an electron.

While the mathematics depends heavily on the reconstruction by Kochen, it makes use of von Neumann $*$-algebras (see Thirring (Thirring 1979)) rather than the Boolean algebras used by Kochen. This choice may seem unnecessarily mathematical for non-relativistic quantum mechanics but it is made to avoid any hint of a logical interpretation and will be useful when discussing relativistic quantum mechanics and field theory. This formulation also differs from Kochen's in the treatment and assignment of both intrinsic and extrinsic properties. The extrinsic properties will be treated here as an actualisation of potential, *dispositional properties*, of the quantum

entity that are shaped by its context. Dispositional properties are represented by operators and contingent properties are not fixed but can take on different values depending on the context of the interaction. The intrinsic properties, such as mass and charge, are fixed and do not change with context.

1.1. Observables and σ-Complexes

In the mathematical formulation of quantum mechanics, a physical system, $\mathbf{S}$, is characterised by a list of properties (called observables in standard quantum mechanics) that can be represented by abstract [1] self-adjoint operators:

$$\mathcal{O}_{\mathbf{S}} = \{\hat{X}_i = \hat{X}_i^* \mid i \in \mathcal{I}_{\mathbf{S}}\},$$

with $\mathcal{I}_{\mathbf{S}}$ a set of indices depending on $\mathbf{S}$, where every operator $\hat{X} \in \mathcal{O}_{\mathbf{S}}$ represents a physical property characteristic of $\mathbf{S}$, such as the total momentum, energy, or spin of all particles belonging to an ensemble of (possibly infinitely many) particles constituting the system $\mathbf{S}$. In standard quantum mechanics there are properties that are not represented by abstract operators, such as charge and mass. In non-formal presentation the space on which operators operate is often left unspecified in strict mathematical terms. In the formal presentation here, the space on which the operators act is a separable Hilbert space $\mathcal{H}_{\mathbf{S}}$ associated with the system $\mathbf{S}$. A system is a related collection of quantum objects.

At every time t, there is a representation of $\mathcal{O}_{\mathbf{S}}$ by self-adjoint operators acting on a separable Hilbert space $\mathcal{H}_{\mathbf{S}}$:

$$\mathcal{O}_{\mathbf{S}} \ni \hat{X}_i \mapsto X_i = X_i^* \in \mathcal{A}(\mathcal{H}_{\mathbf{S}}),$$

[1] For abstract operators the space on which they operate is not specified.

where $\mathcal{A}(\mathcal{H}_{\mathbf{S}})$ is the $*$-algebra of all bounded operators acting on $\mathcal{H}_{\mathbf{S}}$.

In general, the operators in the $*$-algebra do not all commute. This has deep physical consequences:

- Properties that cannot appear together are represented in quantum mechanics by self-adjoint operators that do not all commute.
- The sets of operators that do commute provide a set of properties for a measurable mathematical structure which is a σ-algebra.

The σ-algebra is essential to constructing mathematical probability models using the Kolmogorov axiomisation (Kolmogorov 2018). Therefore, each collection of commuting observables of the object has an associated σ-algebra. So, a collection of σ-algebras that together capture all the property values associated with the object covers all possible events for that object in itself. This structure is the minimal one which contains all the σ-algebras arising from all the properties of the system.

Each dispositional property of an object, such as spin or position, has a σ-algebra of possible values that appear in interaction with other objects. All these σ-algebras form a complex property structure for the object under consideration. The formal definition of this notion is as follows:

> **Definition:** Let F be a family of σ-algebras. The σ-complex Q_F based on F is the union, $\cup B$, of all σ-algebras B lying in F:

$$Q_F = \bigcup_{B \in F} B.$$

Generally, the family F is left implicit, and reference is simply made to a σ-complex Q; however, the family F does tie the complex to an object that has properties that take values depending on the set

of possible interactions that it can have with other objects. Usually, σ-complexes that are closed under the formation of sub-σ-algebras are discussed. It is possible, in any case, to always close a σ-complex by adding all its sub-σ-algebras. This complex will include all the possible values that the dispositional properties of a particle can take in all possible interaction contexts. B denotes the σ-algebra relating to the property values of a particular (commuting) set of observables of the particle or system.

In the mathematical formulation of quantum mechnics, $\mathcal{H}$ is a Hilbert space. For the quantum object under consideration, all properties are represented by self-adjoint operators and possible events are represented by projection operators. For pairwise commuting projection operators, closure under the operation of orthogonal complement and countable product $\prod_i \pi_i$ forms a σ-algebra. The family of all such σ-algebras forms a σ-complex, $Q(\mathcal{H})$, for the object in its possible interaction contexts.

1.2. Proposition

> *In quantum mechanics, the operators representing the possible values of the properties of a system form the σ-complex $Q(\mathcal{H})$ of projections of the Hilbert space $\mathcal{H}$ of the system.*

It has been shown (Kochen 2015) that this proposition holds and the mathematical formalism using $Q(\mathcal{H})$ is equivalent to the standard quantum formulation of the theory on $\mathcal{H}$. However, the interpretation of the formalism here is that while a quantum object has a complete set of properties, they appear only when the situation invokes the σ-algebra in the complex that allows that property to take a specific set of values.

1.3. Quantum States

As presented in Section 1.1, the properties of a quantum object are represented by self-adjoint operators, and these operators act on the elements of a Hilbert space. The concept of the state of a quantum system or object will now be presented. This concept of state is closely aligned with that in statistical mechanics, but the probabilistic quantum state is a property of an individual object rather than a statistical ensemble. As quantum states generalise classical probabilities, the discussion starts with a brief overview of probability measures.

1.3.1. Probability Measures

Classical probability (Kolmogorov 2018) provides a probability measure—a countably additive, $[0, 1]$-valued measure—that is a function:

$$p : B \to [0, 1]$$

with domain B, a σ-algebra of subsets of the state space Ω, such that $p(\mathbf{1}) = 1$, where $\mathbf{1}$ is the identity in the algebra B, and:

$$p\left(\bigcup_i e_i\right) = \sum_i p(e_i)$$

for pair-wise disjoint elements $ e_1, e_2, ...$ in B.

In the case of a system **S**, an object more complex than a single particle, the probability function p gives the probabilities over the σ-algebra of events for that system as a whole. A physical theory predicts the probabilities of outcomes of any possible situation given the complete initial state.

States of a system **S** in quantum mechanics are, in general, represented by density matrices acting on a Hilbert space, $\mathcal{H}_{\mathbf{S}}$. Density matrices are non-negative, trace-class operators, ρ, of trace 1, that is:

$$\rho = \rho^* \geq 0, \quad \text{with } \mathrm{tr}(\rho) = 1.$$

The expectation of a physical quantity, $\hat{X} \in \mathcal{O}_{\mathbf{S}}$ (the set of properties for the system), at time t in a state given by a density matrix ρ is defined by:

$$p(X(t)) := \mathrm{tr}(\rho X(t))$$

where $X(t)$ is the operator representing the physical property $\hat{X}(t)$. This can be extended to all bounded operators in $\mathcal{A}(\mathcal{H}_{\mathbf{S}})$, the $*$-algebra of all bounded operators acting on $\mathcal{H}_{\mathbf{S}}$.

From the mathematical formulation by Kochen (Kochen 2015), adapted here, there is a one-to-one correspondence between states p on the σ-complex $Q(\mathcal{H})$ and density operators.

The use of the same symbol as for the classical event probability $p(x)$ for the quantum state is justified because this mathematical description of the quantum state is a generalisation of the classical probability model from a single σ-algebra to a σ-complex. A density operator ρ defines a probability measure p on $Q(\mathcal{H})$, the σ-complex of projections. The converse, that a state p defines a unique density operator ρ on $\mathcal{H}$, follows from a theorem of Gleason (Gleason 1957). A state on the lattice of projections on $\mathcal{H}$ defines a unique density operator. A lattice consists of a partially ordered set in which every pair of elements has a unique least upper bound and a unique greatest lower bound.

1.4. Pure and Mixed States

In Quantum Mechanics, there is a one-to-one correspondence between the pure states of a system and rays of unit vectors ψ in $\mathcal{H}$, such that $p(X) = (\psi, X\psi)$. The pure states correspond to one-dimensional projections π_ψ (with ψ in the image of π_ψ) and $p(X) = \mathrm{tr}(\pi_\psi X) = (\psi, X\psi)$.

That even the pure states predict probabilities that are not 0 or 1 is what is to be expected of projections onto property values. A pure state simply predicts the probabilities of property values that can become actual in interactions. Mixed states are mixtures of the pure states, and in general, there is no unique decomposition of a mixed state into pure states.

1.5. Properties or Observables

An observable is nothing more than a property of a quantum object that has a set of possible values that become actual values in interaction with other objects and can appear in various contexts. It need not actually be "observed" in an experiment to do so. In the theory, each such property is represented by a self-adjoint operator that has a spectrum that corresponds to the set of possible property values.

This concept of property can, as argued above, be treated within the σ-complex formulation (Kochen 2015). Kochen shows that there is a one-to-one correspondence between observables ω such that:

$$\omega : B(\mathbb{R}) \to Q(\mathcal{H}),$$

where $B(\mathbb{R})$ is the σ-algebra of Borel sets (any set in a topological space that can be formed from open sets, or from closed sets, through the operations of countable union, countable intersection, and relative

complement is a *Borel set*) generated by the open intervals of the real numbers, $\mathbb{R}$.

Given $\pi_\lambda = \omega((-\infty, \lambda])$, there is a Hermitian operator A such that:

$$A = \int \lambda \, d\pi_\lambda .$$

Conversely, given a Hermitian operator A on $\mathcal{H}$, the spectral decomposition above defines the representation of a property ω as the spectral measure $\omega(s) = \int_s d\pi_\lambda$, for $s \in B(\mathbb{R})$. This establishes the one-to-one correspondence and the equivalence of a key element of the reconstruction to that of standard quantum mechanics.

It follows that if $\omega : B(\mathbb{R}) \to Q(\Omega)$ is a representation of a property with corresponding self-adjoint operator X, then, for the state p with corresponding density operator ρ, the expectation of ω is:

$$\mathrm{Exp}_p(\omega) = \mathrm{tr}(X\rho).$$

The result shows the close connection between the spectrum of an operator and the σ-algebra of property values. For instance, for the case of a discrete operator X, the spectral decomposition $X = \sum a_i \pi_i$ defines the σ-algebra of property values generated by the set $\{\pi_i\}_i$. Conversely, given the σ-algebra of property values, the elements of the set $\{\pi_i\}_i$ allow the definition, for each sequence of real numbers a_i, of the Hermitian operator $\sum a_i \pi_i$.

1.6. Symmetries

In the standard formulation of quantum physics, symmetries appear as unitary transformations of Hilbert space vectors or Hermitian operators. Here they also appear naturally as symmetries of a σ-complex.

An automorphism of a σ-complex Q is a one-to-one transformation $\alpha : Q \to Q$ of Q onto Q such that for every σ-algebra B in Q and all $e, e_1, e_2, ...$ in every B:

$$\alpha(e^{\perp}) = \alpha(e)^{\perp} \quad \text{and} \quad \alpha\left(\sum_i e_i\right) = \sum_i \alpha(e_i),$$

where $\perp$ denotes the orthogonal complement.

There is a one-to-one correspondence between symmetries $\alpha : Q(\mathcal{H}) \to Q(\mathcal{H})$ and unitary operators u on $\mathcal{H}$ such that $\alpha(X) = uXu^{-1}$, for all $X \in Q(\mathcal{H})$. If a state p corresponds to the density operator ρ, then:

$$p_{\alpha}(X) = p(\alpha^{-1}(X)) = \mathrm{tr}(\rho u^{-1} X u) = \mathrm{tr}(u\rho u^{-1} X) \tag{1.1}$$

so that the state p_{α} corresponds to the density operator $u\rho u^{-1}$. This shows the mathematical equivalence between the σ-complex and the density matrix formulations under symmetry transformation.

Now that the mathematical formalism in terms of σ-complexes has been presented, it is possible to start to develop a theory that describes the micro-physical world as governed by quantum chance. By chance is meant objective probability that is a physical property of objects in the real sphere rather than a description of uncertainty in the epistemic sphere. This will, among other attributes, allow the derivation of the quantum conditional probability that provides an explanation of quantum state reduction.

In standard quantum mechanics, the term "reduction" usually means the reduction of the wavefunction and so is conceptually tied to the Schrödinger picture. Although there is a mathematical equivalence between the Schrödinger picture and the formalism developed in this paper, the state as a set of probability measures on a σ-complex of projection operators describes a physics of dispositions rather than waves in either real space or some more abstract space. As such, the

position is closer to that of matrix mechanics of early quantum physics, but with more ontological detail on the role of probabilities.

1.6.1. Conditional States

State reduction is a phenomenon that may seem unique to quantum mechanics and has no counterpart in classical mechanics, but this is not the case. Conditional probability plays that role classically, but the objective ensemble or relative frequency interpretations of probability are available, which means that no conceptual problem is posed by the "reduction" of the probability distribution. With this in mind, conditional probability will now be generalised to quantum mechanics.

If p is a state on $Q(\mathcal{H})$ and Y a projection operator in $Q(\mathcal{H})$ such that $p(Y) \neq 0$, then it will be shown that there exists a unique state $p(\cdot \mid Y)$ conditional on Y. If ρ is the density operator corresponding to p, then to see that the operator $Y\rho Y/\mathrm{tr}(Y\rho Y)$ corresponds to the state $p(\cdot \mid Y)$, note that if X lies in the same σ-algebra as Y, then X and Y commute, so:

$$\begin{aligned}\mathrm{tr}(Y\rho YX)/\mathrm{tr}(Y\rho Y) &= \mathrm{tr}(\rho XY)/\mathrm{tr}(\rho Y)\\ &= p(X \cdot Y)/p(Y)\\ &= p(X \mid Y),\end{aligned}$$

which is the standard form for conditional probability. It is proposed that the expression:

$$p(X \mid Y) = \mathrm{tr}(Y\rho YX)/\mathrm{tr}(Y\rho Y) \tag{1.2}$$

generalises the conditional probability state to the case when X and Y do not commute.

The mapping of ρ to $Y\rho Y/\mathrm{tr}(\rho Y)$ is the formula for the reduction of state given by the von Neumann-Lüders Projection Rule (Lüders 1951). In standard quantum mechanics, this rule is an additional principle appended to quantum mechanics. Here it appears as the unique answer to conditioning a state to a specific projection operator.

From the symmetry transformation of the state, equation (Equation 1.1), it follows that the conditional state transforms under a symmetry α as:

$$\begin{aligned} p_\alpha(X \mid Y) &= \frac{p_\alpha(XY)}{p_\alpha(Y)} \\ &= \frac{p(\alpha^{-1}(XY))}{p(\alpha^{-1}(Y))} \\ &= \frac{p(\alpha^{-1}(X)\alpha^{-1}(Y))}{p(\alpha^{-1}(Y))} \\ &= p(\alpha^{-1}(X) \mid \alpha^{-1}(Y)) \end{aligned}$$

The results in this subsection hold when X and Y are in the same commutative von Neumann sub-algebra corresponding to a sub-σ-algebra of $Q(\mathcal{H})$. The next subsection covers the more interesting case of when they are in different sub-algebras and there is a correction to the form of the conditional probability that is well known from classical probability.

1.6.2. Total Probability in Classical and Quantum Conditional Probability

The Law of Total Probability (or Law of Alternatives in the discrete case, as here) provides a useful method to partition and analyze conditional probabilities. Firstly, in the classical case, let $Y_1, Y_2, ...$ lie in a $*$-algebra of commuting projection operators with $Y_i Y_j = 0$ for $i \neq j$, and let $Y = \sum_i Y_i$, then:

$$\begin{aligned} p(X \mid Y) &= p\left(\sum_i (XY_i)\right)/p(Y) \\ &= \sum_i (p(XY_i)/p(Y_i))(p(Y_i)/p(Y)) \\ &= \sum_i p(X \mid Y_i)p(Y_i \mid Y) \end{aligned}$$

Not only is this partition useful, but it also displays clearly how the quantum case differs from the classical.

From Equation 1.2 above:

$$\begin{aligned} p(X \mid Y) &= \mathrm{tr}(Y\rho YX)/\mathrm{tr}(\rho Y) \\ &= \mathrm{tr}\left(\sum_{i,j} Y_i\rho Y_j X\right)/\mathrm{tr}(\rho Y) \\ &= \sum_i \mathrm{tr}(Y_i\rho Y_i X)/\mathrm{tr}(\rho Y) + \sum_{i\neq j} \mathrm{tr}(Y_i\rho Y_j X)/\mathrm{tr}(\rho Y) \\ &= \sum_i p(X \mid Y_i)p(Y_i \mid Y) + \sum_{i\neq j} \mathrm{tr}(Y_i\rho Y_j X)/\mathrm{tr}(\rho Y). \end{aligned}$$

This shows that for conditional properties, an interference term must be added to the classical law of total probability. When X commutes with each Y_i, the interference term is zero.

A measurement of a property is the most familiar example of conditioning with respect to several properties. If a property represented by an operator A has a spectral decomposition $\sum_i a_i\pi_i$, then measuring the observable amounts to registering the values of the properties given by the π_i. The interaction algebra B_A is that generated by the π_i.

Conditioning can be used to define the context for the dynamics of a quantum object. In general, this requires the notion of conditioning with respect to several conditions.

Given a system with σ-complex Q and disjoint elements $Y_1, Y_2, ...$ in a common σ-algebra in Q with $\sum_i Y_i = \mathbf{1}$, and a state p, the state conditional on $Y_1, Y_2, ...$ is defined to be $p(\cdot \mid Y_1, Y_2, ...) = \sum p(Y_i)p(\cdot \mid Y_i)$. This can be written more compactly as $p(\cdot \mid B_A)$, the state conditional on the interaction algebra B_A.

For a quantum system, with σ-complex $Q(\mathcal{H})$ and if ρ is the density operator corresponding to the state p:

$$p(\cdot \mid B_A) = \sum \mathrm{tr}(\rho Y_i)(Y_i \rho Y_i / \mathrm{tr}(\rho Y_i)) = \sum Y_i \rho Y_i,$$

so that for each X the probability $p(X \mid B_A) = \sum_i \mathrm{tr}(Y_i \rho Y_i X)$.

The natural definition for applying a symmetry to the conditioned state is given by:

$$p_\alpha(X \mid B_A) = p(\alpha^{-1}(X) \mid \alpha^{-1}(B_A)).$$

The σ-algebra B_A generated by the $Y_1, Y_2, ...$ is simply a sub-algebra of the σ-complex.

There is an equivalence between quantum states and the probabilities describing objective chance. The reconstruction provides a generalisation of probability theory that is made evident in the equation for the conditional probability that introduces extra terms due to quantum interference.

1.7. The Measurement Problem

This treatment is adapted from Kochen's reconstruction of quantum mechanics (Kochen 2015). It is a mathematical formulation of the theory that is equivalent to the standard formulation, but it is not a physical theory. The physical theory is the interpretation of the mathematics. The interpretation here is that the properties of a quantum

object appear only in interaction with other objects. The measurement problem arises because the standard formulation of quantum mechanics assumes that an isolated system evolves unitarily according to Schrödinger's equation, but when a measurement occurs, an eigenvalue of the operator representing the observable being measured is randomly selected as the result of the measurement. This additional mechanism is not explained by the unitary evolution and leads to a contradiction between the predicted state and the observed state after measurement.

The Measurement Problem refers to a postulate in standard quantum mechanics, which assumes that an isolated system undergoes unitary evolution via Schrödinger's equation and then an eigenvalue of the operator representing the observable being measured (an observable is a property of the system that the experimental setup is designed to measure) is randomly selected as the result of the measurement, as presented by Bohm (2001), for example. However, if a property $\hat{A}$ of a system S is measured by an apparatus T, the total system $S + T$, if assumed to be isolated, then undergoes unitary evolution. The random selection of an eigenvalue is an additional mechanism.

1.8. Ideal Measurement

The mathematical formulation of an ideal measurement, in standard quantum mechanics, is as follows for system S in a pure state ϕ_k:

Take the spectral decomposition of an operator representing an observable to be

$$A = \sum_i a_i \pi_i.$$

Each π_i is a one-dimensional projection with eigenstate ϕ_i and $\{a_i\}_i$ is the set of eigenvalues. The apparatus T is assumed to be sensitive to the different eigenstates of A.

Hence, if the initial state of S is ϕ_k and the apparatus T is in a neutral state ψ_0, so that the state of $S+T$ is

$$\phi_k \otimes \psi_0,$$

the system evolves into the state

$$\phi_k \otimes \psi_k,$$

where the $\{\psi_i\}_i$ are the states of the apparatus operator corresponding to the states $\{\phi_i\}_i$ of the system S.

T and its interaction with S will have been chosen to achieve this: a perfectly designed measurement apparatus to be in ψ_l whenever S is in ϕ_l for all l.

But now, for the case of a more general initial state

$$\phi = \sum_i c_i \phi_i,$$

linearity gives:

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i.$$

A problem with this for standard quantum mechanics is that the completed measurement gives a particular apparatus state ψ_k, say, indicating that the state of S is ϕ_k, so that the state of the total system is $\phi_k \otimes \psi_k$, in contradiction to the derived evolved state

$$\sum_i c_i \phi_i \otimes \psi_i.$$

This evolution does not describe what happens in an experiment.

1.9. Conditioning and Reduction

In contrast, the reduction can also be considered from the viewpoint of the conditioning of the states. If the state p of $S+T$ just prior to measurement is ρ_Γ, corresponding to

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i,$$

then after the measurement it is in the conditioned state, by Equation 1.2:

$$\begin{aligned} & p(\cdot \mid (\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})) \\ &= \frac{(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\rho_\Gamma(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})}{\mathbf{tr}((\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\rho_\Gamma)} \\ &= \frac{(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\left(\sum_i c_i \phi_i \otimes \psi_i\right)(\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})}{\mathbf{tr}((\pi_{\phi_k} \otimes I)(I \otimes \pi_{\psi_k})\left(\sum_i c_i \phi_i \otimes \psi_i\right))} \\ &= \pi_{\phi_k \otimes \psi_k}. \end{aligned}$$

Hence, the new conditioned state of $S+T$ is the reduced state $\phi_k \otimes \psi_k$. This is not surprising as it is conditioned on being in just the kth state of ρ_Γ and so it projects that element out. This is not a resolution of the measurement problem but merely makes use of the probabilistic formulation of the theory to show conditioning forces the state reduction.

1.10. Beyond Unitary Evolution

Whereas standard quantum mechanics must add a means to reconcile the unitary evolution of $S+T$ with the measured reduced states of S and T, the interpretation argued for in this paper takes the opposite approach to the orthodox interpretation. The point of departure is not

the unitary development of an isolated system, but rather the result of an interaction.

It is the conditions under which dynamical evolution occurs that must be further investigated, rather than the additional reduced state mechanism. Therefore, it should not be taken for granted, as assumed in standard quantum mechanics, that an isolated system evolves unitarily.

The question to be addressed is whether in a measurement the σ-complex structure of $S+T$ undergoes a symmetry transformation at separate times of the process. This is formalised as the condition for the existence of a representation

$$\alpha : \mathbb{R} \to \mathrm{Aut}(Q).$$

The outcome of a measurement cannot be given by a unitary process.

1.11. Failure of Automorphism Description

A completed measurement or a state preparation has two distinct elements of

$$Q(\mathcal{H}) = Q(\mathcal{H}_S \otimes \mathcal{H}_T)$$

at initial time 0 which end up being mapped to the same element at a later time t.

One such element is an initial state $\phi \otimes \psi_0$ resulting in a state $\phi_k \otimes \psi_k$, for some k. However, a second such element $\phi_k \otimes \psi_0$ also results in the state $\phi_k \otimes \psi_k$.

If the state ϕ is chosen to be distinct from ϕ_k, then the two elements

$$\pi_{\phi\otimes\psi_0}, \qquad \pi_{\phi_k\otimes\psi_0}$$

of $Q(\mathcal{H})$ both map to the same element

$$\pi_{\phi_k \otimes \psi_k}.$$

However, any automorphism α_t is a one-to-one map on Q, so the measurement process cannot be described by a representation

$$\alpha : \mathbb{R} \to \mathrm{Aut}(Q),$$

and hence **a unitary evolution cannot explain what is observed**.

1.12. State Reduction in Dynamics

The Measurement Problem must be resolved by a theory that includes state reduction in its dynamics in addition to periods of unitary evolution. The GRW theory provides an example of a partial mechanism for this (Ghirardi et al. (1986)) — partial because it only reduces the wavefunction to one that is more localised rather than full transition from possibility to actuality.

However there is a nonlocal guiding field that is problematic as a spacetime entity.

Kochen's reconstruction does not provide a non-unitary reduction mechanism. This will require a richer theory. The only current candidate is Relativistic Transactional Interpretation (RTI) (Kastner (2022)), which is a relativistic extension of the Transactional Interpretation of quantum mechanics (Cramer (1986)). RTI provides a mechanism for state reduction that is consistent with relativity and does not require a preferred frame of reference.

1.13. Internal Interactions

For a composite system it should not only be outside forces that can break symmetry, but internal interactions. In the state

$$\Gamma = \sum_i c_i \phi_i \otimes \psi_i$$

introduced above, the total, but still isolated, system $S + T$ has a set of i property values associated with the states $\{\phi_i \otimes \psi_i\}_i$.

However, the interacting object T as part of the system $S + T$ will have the state ϕ_k of S appear with probability $|c_k|^2$. This provides a matrix mechanics interpretation of reduction as a physical transition probability for the system S in the presence of the apparatus T.

State reduction does take place in isolated compound systems with internal interactions and the reduction of the state is due to the combined system's properties but traceable to the dispositional power to take specific property values associated with S.

1.14. The double slit experiment

Having discussed the issues with quantum measurement in general, and shown that standard quantum mechanics is incomplete even in a probabilistic reconstruction, Young's double slit experiment with electrons will be discussed in two variants (for background see Chapter 1 of Hall (2013)): The standard configuration, to be described below, Figure 1.1. A configuration with a pointer that acts behind the slits to point in the direction of the passing electron, Figure 1.2.

1.14.1. Standard configuration

The experiment, Figure 1.1, is as follows:

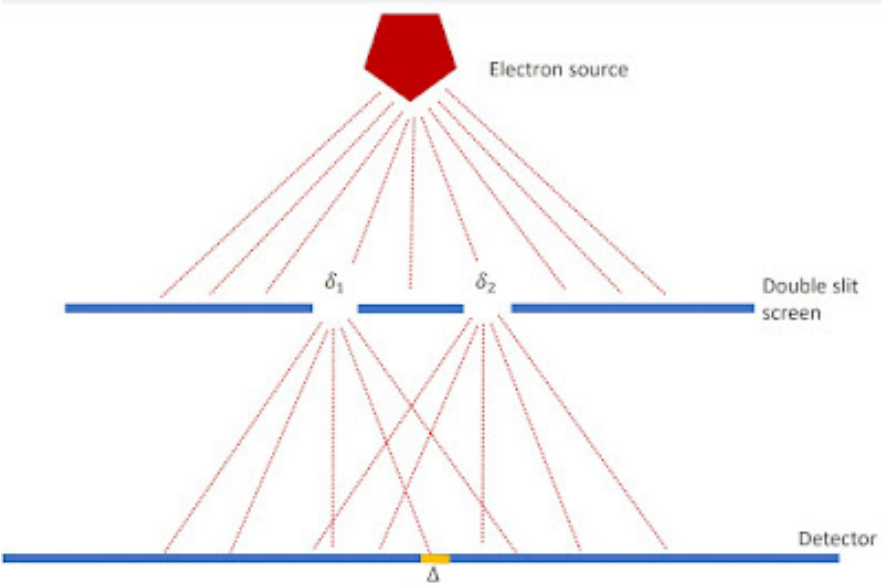


Figure 1.1.: The setup for the double slit experiment is shown. An electron source sends one particle at a time toward the screen with the slits. The slits are marked by δ_1 and δ_2 and a sample region Δ is shown on the detector screen.

There is a source of electrons that move towards a screen with two slits.
The intensity of the beam is low and only one electron is moving towards the detector at any time.
The slits are marked δ_1 and δ_2.
Δ be some arbitrary region on the electron detector.

Let Y_1 and Y_2 be the projection operators for position in the regions of the two slits δ_1 and δ_2. Then $Y_1 + Y_2$ is the projection of position for the union $\delta_1 \cup \delta_2$. Let X be the operator for position in a local region Δ on the detection screen. Assume the electron is only constrained to pass through the slits without being constrained as to which, then under those conditions the conditional probability is given by the Law of Equation 1.2:

$$p(X|Y_1 + Y_2) = p(X|Y_1)p(Y_1|Y_1 + Y_2) + p(X|Y_2)p(Y_2|Y_1 + Y_2)$$
$$+\frac{\mathrm{tr}(Y_1\rho Y_2 X) + \mathrm{tr}(Y_2\rho Y_1 X)}{\mathrm{tr}((Y_1 + Y_2)\rho(Y_1 + Y_2))}.$$

where the property of projection operators, $PP = P$ has been used. This can be written more compactly as

$$p(X|Y_1 + Y_2) = p(X|Y_1)p(Y_1|Y_1 + Y_2)$$
$$+p(X|Y_2)p(Y_2|Y_1 + Y_2) + p(X|Y_1 + Y_2)_I$$

where $p(X|Y_1 + Y_2)_I$ is the interference term.

Note that if X commutes with either Y_1 or Y_2, this interference term vanishes, because $Y_i Y_j = 0$ for $i \neq j$. This can be observed if the detector is right next to the two-slit screen because X then coincides with either Y_1 or Y_2. If the detector is a distance from the two-slit screen (e.g. $X = \Delta$), then the state of the electron evolves unitarily

via α_t, so $\alpha_t(Y_i) = u_t Y_i u_t^{-1}$ no longer commutes with $\alpha(X)$, giving rise to the non-zero interference term.

The standard explanation of the interference effect is that the state of the particle is, or acts as, a coherent pair of waves emanating from the slits, which exhibit constructive and destructive interference effects. This was, of course, the explanation for Young's original experiment with light. For individual quantum particles, however, there is the unexplained local event observed at the detection screen. This is a problem for the theory. While the Born interpretation of the wavefunction provides a probability distribution for the particle position it requires the detection screen, operating outside what is described by the mathematical theory, to act as a sampling mechanism for that distribution.

The explanation given in this blog is that the two-slit screen functions as a preparation of the state for the particle, by which the state is conditioned, or reduced, to pass through the region $\delta_1 \cup \delta_2$. This reduction is not a position measurement, since $\delta_1 \cup \delta_2$ is not a localised region (as it would be for a single-slit screen). Once the particle reaches a detection screen then, in interaction with the screen, it appears in a random local region Δ and its position takes a value. Just as in the standard Born interpretation of the wavefunction, it is not explained in the theory how the electron takes the value that the detector detects other than invoking random sampling of the possible values.

So, the interference pattern on the detector is built up over time as more electrons arrive and are sampled by the detector.

1.14.2. The introduction of an interaction with a pointer

This section is adapted from Bricmont (2017), Appendix 5.A and Maudlin (2019).

The experiment, illustrated in Figure (2), is now as follows:

There is again a source of electrons that move towards a screen with two slits.
The intensity of the beam is low and only one electron is moving towards the detector at any time.
The slits are marked δ_1 and δ_2.
Δ be some arbitrary region on the electron detector.

A pointer P is introduced. It is a quantum object with three states: neutral P_0, points to slit 1 (P_1), and points to slit 2 (P_2). The interaction with the electron causes the pointer to move towards it.

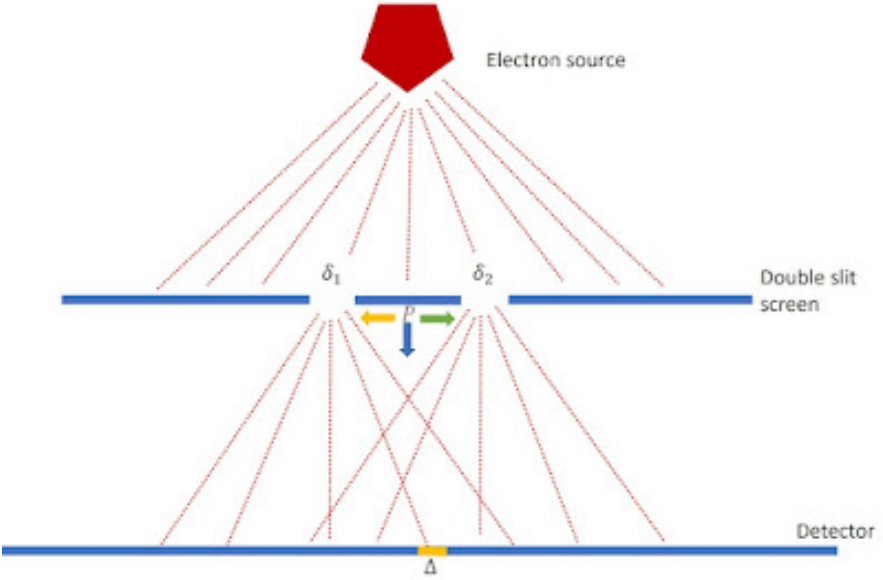


Figure 1.2.: The setup for the double slit experiment with a pointer behind the slits is shown.

Again let Y_1 and Y_2 be the projection operators for position in the regions of the two slits δ_1 and δ_2. Then $Y_1 + Y_2$ is the projection of

position for the union $\delta_1 \cup \delta_2$. Let X be the operator for position in a local region Δ on the detection screen. The operator representing the pointer has three eigenstates and therefore a three-dimensional Hilbert space $\mathcal{H}_P$. Without the pointer the Hilbert space is $\mathcal{H}_0$. The Hilbert space of the total system is $\mathcal{H}_P \otimes \mathcal{H}_0$.

The possible constituent states are:

- ϕ_1 pointer pointing to slit δ_1
- ϕ_2 pointer pointing to slit δ_2
- ϕ_0 pointer neutral
- ψ_1 electron through slit δ_1
- ψ_2 electron through slit δ_2
- Ψ_0 electron with pointer in neutral state P_0

Assuming the pointer starts in its neutral state, the initial wave function is

$$\begin{aligned}\Psi_0 &= \phi_0 \otimes (\psi_1 + \psi_2) \\ &= \phi_0 \otimes \psi_1 + \phi_0 \otimes \psi_2\end{aligned}$$

1.14.2.1. Treatment I: Pointer reacts to which slit the electron passes through

Time evolution gives:

$$\Psi = \phi_1 \otimes \psi_1 + \phi_2 \otimes \psi_2.$$

The interference term becomes:

$$p(X|Y_1 + Y_2)_I = \frac{\mathrm{tr}(Y_1 \rho_\Psi Y_2 X) + \mathrm{tr}(Y_2 \rho_\Psi Y_1 X)}{\mathrm{tr}(\rho_\Psi (Y_1 + Y_2))}$$

Using orthogonality of pointer states:

$$p(X|Y_1 + Y_2)_I = 0$$

Interference term disappears.

1.14.2.2. Treatment II: Pointer does *not* react to which slit the electron passes through

General evolution:

$$\Psi = \sum_{i \in \{0,1,2\}} a_i \phi_i \otimes \psi_1 + \sum_{i \in \{0,1,2\}} b_i \phi_i \otimes \psi_2$$

with

$$a_0 = b_0, \quad a_1 = b_2, \quad a_2 = b_1.$$

The detection pattern is:

$$(\Psi, P \otimes X\, \Psi) = \sum_i |a_i|^2 (\psi_1, X\psi_1)$$

$$+ \sum_i |a_i|^2 (\psi_2, X\psi_2) + \sum_i a_1^* a_2 (\psi_2, X\psi_1) + \sum_i a_2^* a_1 (\psi_1, X\psi_2)$$

Let $C = \sum_i |a_i|^2$:

$$(\Psi, P \otimes X\, \Psi) = C\big[(\psi_1, X\psi_1) + (\psi_2, X\psi_2)\big] + \mathrm{Re}\{2a_1^* a_2 (\psi_2, X\psi_1)\}.$$

The interference term $\mathrm{Re}\{2a_1^* a_2 (\psi_2, X\psi_1)\}$ is non-zero, so the interference pattern is present if an experiment is carried out.

1.15. Critique of Experimental Interpretation

The results of the double slit experiment are often interpreted as showing that the electron is a wave that passes through both slits and interferes with itself. However, this interpretation is not supported by the formalism of quantum mechanics. The formalism only provides probabilities for the outcomes, and does not describe the behaviour of the electron between these outcomes. The analysis of the double slit experiment shows that the interference pattern arises from the conditional probabilities of the electron's position given the slits, and that the presence of a pointer that interacts with the electron can eliminate this interference. This suggests that the electron does not have a definite path through the slits, but rather that its behaviour is influenced by the total experimental context. The discussion of the double slit experiments above is solely in terms of probabilities. That is, in terms of potential events. Standard quantum mechanics has no explanation of how to go from potential events to actual events. The theory provides probabilities for the outcomes of measurements, but does not describe the process by which these outcomes are realised. This deficiency remains in the probabilistic reconstruction of quantum mechanics. The theory provides a framework for understanding the relationships between different properties and their probabilities, but it does not provide a mechanism for the actualisation of these properties in specific experimental contexts.

In an experiment the results are recorded at the time of the experiment. This experimental recording is not part of the formal theory. The theory provides transition probabilities but nothing to time the transition. To address this, a more complete theory is needed that includes the dynamics of state reduction and the actualisation of properties in specific contexts. The Relativistic Transactional Interpretation (Kastner (2022)) is one such candidate that aims to provide a mechanism for state reduction consistent with relativity, allowing for a more complete understanding of how potential events become actual events in quantum mechanics. This will be discussed in the next Chapter 4 where the Relativistic Transactional Interpretation is presented in the generalised probabilistic form outlined in this chapter. This will address the difficulties inherent in standard quantum mechanics and its probabilistic reconstruction.

Part II.

Locality and Realism

2. Locality, Microcausality, and Entanglement

Until now, the discussion has tried, as far as possible, to free quantum mechanics from reference to measurement but quantum mechanics also has a problem with locality. First, it is worth remembering that classical mechanics also had a locality problem and it was tolerated for two centuries. Newton's theory of gravity was published in 1687 and structurally similar Coulomb's law of electric charge attraction and repulsion was published in 1785. In both cases any local change in mass or charge, whether magnitude or position, had an instantaneous effect everywhere. There was no mechanism in the physics for propagation of the effect of a change on mass (gravity) or charge (electricity). A local change had an effect across the universe. The solution to this was found first for electricity in combination with magnetism. Faraday proposed the existence of a field. The mathematical formulation of this concept by Maxwell led to the classical electromagnetic theory (1865) and provided a propagation mechanism.

The success of electromagnetic theory brought to the fore locality problem with classical dynamics. The space and time translation invariance in classical Newtonian dynamics did not follow the same transformation rules as in the electromagnetic theory and there was still no mechanism proposed for the propagation of gravitational effect. As is well known, Einstein, inspired by Maxwell, solved both anomalies with first the special theory , which indicated that causal effects can not propagate faster than the speed of light in empty space, and then the general theory of relativity that provided a propagation mechanism by

making space-time an active physical entity supporting gravitational waves.

By the time the general theory of relativity was formulated it was evident that classical theory had a further deep problem; it could not explain atomic and other micro phenomena. To tackle this problem solutions were found for specific situations. Max Planck introduced his constant $\hbar$ to resolve the problem of the ultraviolet singularity in the black body radiation spectrum by proposing energy quantisation (Bohm 1951). This same constant came to be fundamental in explaining atomic energy levels, the photoelectric effect role and more generally the quantisation of action.

Quantum theory took shape in the 1920s with the rival formations, by Heisenberg and Schrödinger (with much help from others), shown to be formally equivalent. The space time translation symmetry of special relativity was also built into an equation for the electron proposed by Dirac that in turn implied the existence of anti-matter. But a fully relativistic quantum mechanics remains a research topic (Haag 1996).

To combine particle theory with electromagnetism quantum electrodynamics was developed. This theory was remarkably successful in its empirical confirmation but relied on some dubious mathematical manipulation. To deal with this the mathematical foundations of quantum field theory were examined. It is at this point that the first type of locality to be considered appears in quantum theory in mathematically precise form (Haag 1996).

2.1. Causal Locality

A basic characteristic of physics in the context of special relativity and general relativity is that causal influences on a Lorentzian manifold spacetime propagate in timelike or light-like directions but not space-like. Space-like separated points in space-time lie outside each other's

light cone, which means that no influence can pass from one to the other.

A further way of considering causality is that influences only propagate into the future in time-like and light-like directions, but this is not simple to deal with in either classical special relativity or standard quantum mechanics because of their time reversal symmetry. One approach would be to treat irreversible processes through coarse-grained entropy in statistical physics. But this seems more like a mathematical trick that treats irreversibility as due to a lack of access to the detailed microscopic reversible dynamics. That is, irreversibility is an illusion. A more fundamental approach is to develop a new physics, as is being attempted by Fröhlich (2021) and here in this book.

To return to Einstein causality, any two space-like-separated regions of spacetime should behave like independent subsystems. This causal locality is, with a slightly stronger technical definition, Einstein causality. This concept of locality when adopted in relativistic quantum theory (algebraic theory) implies that space-like separated local self-adjoint operators commute. This is sometimes known as microcausality. Microcausality is causal locality at the atomic level and below.

In quantum theory, where operators represent physical quantities, the microcausality condition requires that any operators commute that pertain to two points of space-time if these points cannot be linked by light propagation. This commutation means, as in standard quantum mechanics, that the physical quantities to which these operators correspond can be precisely determined locally, independently, and simultaneously. However, the operators in standard quantum theories and the non-relativistic alternatives discussed so far in this book don't have a natural definition of an operator that is local in space-time. For example, the position operator is not at any point in space. The points in space are held as potential values in the quantum state that is represented mathematically by the density matrix. How these potential values become actual is dealt with in standard quantum mechanics by the Born criterion, which is, however, tied to measurement situations.

To remove this dependence on measurement situations is a major aim of this book and the measurement only need be invoked when discussing how various form of locality and non-locality are known about.

As the introduction of classical fields cured Newtonian dynamics of action at a distance and eventually modified then replaced Newton's theory of gravity with General Relativity, the development of quantum field theory could perhaps cure standard quantum mechanics of its causal locality problem. Local quantum theory as set out in the book by Haag (1996) tackles this challenge. The technical details involved will be simplified here.

Although dealing with these questions coherently within non-relativistic quantum theory is not mathematically valid, it is instructive to explore specific examples with an emphasis on physical intuition rather than mathematical rigour. Following Fröhlich (2023), it is natural to consider the spin of the particle to be local to that particle. Therefore, the spin operators, whether represented by Pauli matrices or by projection operators that project states associated with some subsets of the spin spectrum, can be assigned unambiguously to one particle or another.

In a situation where two particles are prepared so that they propagated in opposite directions their local interactions with other entities will eventually be space-like separated. The spin operators of one commute with the spin operators of the other. The local interaction of one cannot then be influenced by the local interaction of the other. This is a specific example of microcausality.

But what if the preparation of the two particles entangles their quantum states? This entanglement may persist over any subsequent separation, if the particle does not first undergo any interaction with other particles or fields.

Note that entanglement is a state property whereas microcausality is an operator property and proceed to a discussion of entanglement and its consequences in a developed version of the two-particle example. If algebraic quantum field theory is imposed a similar microcausality

condition on the quantum state, it would be contradicted by the experiments that show the violation of Bell's inequality, see Chapter 3.

The Relativistic Transactional Interpretation (RTI) of quantum mechanics, as discussed in Chapter 4 as Possibilist Transactional Quantum Theory (PTQT), provides a way to understand how entanglement can persist over space-like separations without violating microcausality. In PTQT the operators do not act in space-time so the locality condition discussed above does not apply. The operators act in the quantum substratum, which is a pre-spacetime domain. The incompatibility of the microcausality condition with PTQT will be discussed in Section 4.6.

2.2. Entanglement and Non-Locality

The two-particle example only needs the introduction of a local spin measurement mechanism for each particle for it to become the version of the Einstein Podolsky, Rosen thought experiment formulated by Bohm (1951). This section will follow Bohm's mathematical treatment closely but will avoid as far as possible invoking the results of measurements. Bohm's discussion follows the Copenhagen interpretation but also uses the concept of potentiality as developed by Heisenberg (1958).

The system in this example consists of experimental setups (described below) for two atoms (1 and 2) with spin 1/2 (up/down or $\pm\hbar$). The z direction spin aspect of the state of the total system consists of four basic wavefunctions:

$$\psi_a = \phi_{+,z,1} \otimes \phi_{+,z,2}$$

$$\psi_b = \phi_{-,z,1} \otimes \phi_{-,z,2}$$

$$\psi_c = \phi_{+,z,1} \otimes \phi_{-,z,2}$$

$$\psi_d = \phi_{-,z,1} \otimes \phi_{+,z,2}$$

it will be shown below that although the choice of the z direction is convenient the results of the analysis do not depend on it.

If the total system is prepared in a zero-spin state, then it can be represented by the linear combination:

$$\psi_0 = \psi_c - \psi_d \tag{2.1}$$

This correlation of the spin states of the particles is an example of quantum entanglement.

Each particle also has associated with its spin state a wavefunction that describes its motion and position. These space wavefunctions will not be shown explicitly here but are important conceptually because the particles always move away from each other. The description of the thought experiment is completed by each particle undergoing a Stern-Gerlach experimental interaction at space-like separated regions of space-time, as shown below.

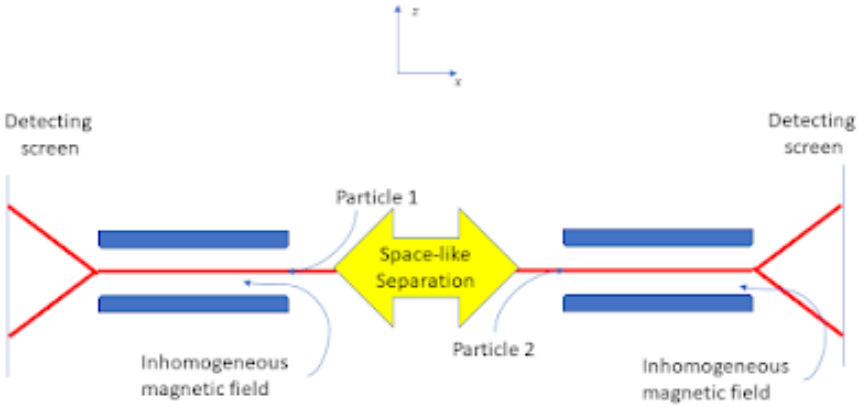


Figure 2.1.: Two space-like separated Stern-Gerlach interaction situations.

The detecting screen is a part of an experimental setup that is needed for confirming the predictions of the theory but not the physics of the effects. Of primary interest is the interaction of the particles with the magnetic field $\mathfrak{H}$. The component of the system Hamiltonian for the interaction of the particle spin with the magnetic field is, from Bohm (1951),

$$\mathcal{H} = \mu(\mathfrak{H}_0 + z_1\mathfrak{H}_0')\sigma_{1,z} + \mu(\mathfrak{H}_0 + z_2\mathfrak{H}_0')\sigma_{2,z}$$

where $\mu = \frac{e\hbar}{2mc}$, $\mathfrak{H}_0 = (\mathfrak{H}_z)_{z=0}$ and $\mathfrak{H}_0' = (\frac{\partial \mathfrak{H}_z}{\partial z})_{z=0}$. m and e are the electron mass and charge. c is the speed of light in vacuum. Assume

the magnetic fields have the same strength and spatial form in both regions but this not essential. It is also assumed that each particle interacts with its own local magnetic field at the same time. This is not a limiting assumption, but it is essential to assume that the time of the interaction is short enough for the local space-time regions to remain space-like separated.

The Schrödinger equation can now be solved for a wavefunction of the form

$$\psi = f_c\psi_c + f_d\psi_d$$

with initial conditions given by Equation 2.1. The result is, once the particles have passed through the region with non-zero magnetic field strength:

$$f_c = \frac{1}{\sqrt{2}}e^{-i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}$$

and

$$f_d = -\frac{1}{\sqrt{2}}e^{i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}$$

Where Δt is the time it takes for the particles to pass through the magnetic field.

Inserting the above results into the equation for ψ gives the wavefunction post interaction:

$$\psi = \frac{1}{\sqrt{2}}e^{-i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}\psi_c - \frac{1}{\sqrt{2}}e^{i\frac{\mu\mathfrak{H}_0'}{\hbar}(z_1-z_2)\Delta t}\psi_d$$

Therefore, for a system prepared with total spin zero undergoing local interactions in space-like separated regions, as shown in the figure,

there is equal probability for each particle to be deflected either up or down. However, because of the correlation when one is deflected up the other is deflected down.

This may seem unsurprising because the total spin is prepared to be zero. No more surprising than taking a green card and a blue card, putting them in identical envelopes, shuffling them and then giving one to a friend to take far away. Opening the envelope you kept and seeing a green card means that the distant envelope contains a blue card. This, clearly, is not a non-local influence.

However, this is not the end of the story. As mentioned above, there is nothing special about the z direction of spin. The same analysis can be carried with x direction states, as follows:

$$\begin{aligned}\psi_{a'} &= \phi_{+,x,1} \otimes \phi_{+,x,2} \\ \psi_{b'} &= \phi_{-,x,1} \otimes \phi_{-,x,2} \\ \psi_{c'} &= \phi_{+,x,1} \otimes \phi_{-,x,2} \\ \psi_{d'} &= \phi_{-,x,1} \otimes \phi_{+,x,2}\end{aligned}$$

and again, the zero total spin state is:

$$\psi_{0'} = \psi_{c'} - \psi_{d'} \tag{2.2}$$

Using the standard spin state relations (valid for both particles one and two, by introducing the appropriate tags (1 or 2), see Bohm (1951):

$$\phi_{+,x} = \frac{1}{\sqrt{2}}(\phi_{+,z} + \phi_{-,z}) \quad \text{and} \quad \phi_{-,x} = \frac{1}{\sqrt{2}}(\phi_{+,z} - \phi_{-,z})$$

Inserting into Equation 2.2, with some algebra, it can be shown that:

$$\psi_{0'} = \psi_0$$

Therefore, if the Stern-Gerlach setup is rotated to measure the x component of spin, exactly the same analysis can be carried out as for the z component giving the same anti-correlation effect. It must be stressed that physical effects are of primary importance and not the results of experiments or the experimenter's knowledge of events at this point, which are secondary.

In general, there is no reason for the two space-like separated setups to be chosen in the same direction. If the choice is effectively random then when the direction of interaction does not coincide there will be no correlation between the outcomes but if they happen to be in the same direction, then there will be the $\pm$ anti-correlation. Locally the spin operators for the x, y and z do not commute. Their values are potential rather than actual and remain non-actual after the interactions. The situation is not like the classical coloured cards in envelopes example. There is no direction of spin fixed by the initial state preparation. Indeed, that would be inconsistent with a total spin zero state preparation. What the interaction does is chose a σ-algebra from the local σ-complex but the spin state of the system remains entangled.

As far as local effects are concerned, each particle behaves as expected for a spin 1/2 particle. This is causal locality. It is only if someone gets access to a sequence of measurements from both regions (here is the only place where detection enters this description of the physics of this situation) that the anti-correlation effect can be confirmed.

The effect depends on the preparation of the initial total system state. There is persistent correlation across any distance just as in the green and blue card example, but it is mysterious because the initial state does not hold an actual value of each spin component for each particle, unlike the actuality green and blue card example. There is no way for the one particle to be influenced by the choice of direction of measurement at the region where the other particle is, but a correlation of potentiality persists that depends on the details of the total quantum state.

It is perhaps too early to simply accept that there are non-causal, non-classical correlations of potentialities between two space-like separated regions. That would be a quantum generalisation of the blue and green card example. What the theory does predict is that the effect due to entanglement is not just epistemic but physical once potentiality is accepted as an aspect of the ontology.

3. Bell's Theorem

Bell's theorem has an untidy history. It is even debatable whether it merits being called a theorem. However, it is extremely important, as shown below, and ought to be shaped into the form of a theorem.

The proof is in two parts, as in Myrvold et al. (2020):

- Formulation of a measurement framework and then derive a version of the inequality;
- demonstrate that quantum mechanics predicts violations of the inequality.

This will show a conflict between a statistical analysis of possible experiments that assumes a locality concept called Bell locality (BL), and textbook quantum mechanics (QM). Further discussion is then devoted to the challenge of deciding experimentally between BL and QM.

The framework is a generalisation of the one used in Bell (1964). Bell's 1964 derivation assumed an experiment involving perfect anticorrelation of the results of aligned Stern-Gerlach experiments on a pair of entangled spin-$\frac{1}{2}$ particles. This is not realistic, and experimental tests need this assumption to be relaxed because of the practical difficulties of maintaining quantum coherence and guaranteeing spacelike separation of the measurements. An experimental setup with polarised photons is more appropriate. Papers that took the steps from Bell's 1964 demonstration to an experimentally testable form include John F. Clauser et al. (1969), Bell (1971), Clauser and Horne (1974), Aspect (1983) and Mermin (1986).

The initial part of the proof will use a framework describing the general experimental setup and the assumptions sufficient to ensure satisfaction of the Bell inequalities. The second part will show that quantum mechanics predicts violations of the inequalities. The conceptual framework postulates an ensemble of configurations of two correlated subsystems, with each entangled system labelled 1 and 2. Each entangled system is characterised by a configuration state z that contains the entirety of its properties, including the properties of the individual sub-systems and any correlations between them. The state z is taken to be a complete specification of the properties of the configuration at the moment of generation and future evolution, other than future experimental decisions.

The state space Z is the totality of all states z, and it is assumed that the subsets of Z are measurable subsets, forming a measurable space to which probabilistic considerations may be applied. The probability distribution over Z is denoted by δ. In addition, it is assumed that the mode of generation of the pairs establishes a probability distribution p that is independent of the histories of each of the two systems after they separate. Temporal evolution of the properties of the two systems after separation is possible, but it is assumed that the state z sets the probabilities for subsequent events (including any temporal evolution), and therefore the probabilities for outcomes of experiments to be performed on the sub-systems separately.

Different experiments may be performed on each subsystem. Use $a \in \mathcal{A}$ as variables ranging over possible experiment settings $\mathcal{A}$ on subsystem 1, and $b \in \mathcal{B}$ as variables ranging over experiment settings $\mathcal{B}$ on subsystem 2. It is not assumed that these parameters capture the complete state of the experimental apparatus, which might have a range of microstates corresponding to each experimental setting; consider them as labels. The following assumption concerns experimental procedures and the independence of the preparation of the entangled system from the choice of experiments to be performed.

Measurement Independence Assumption: the prepa-

ration probability distribution δ is independent of the processes by which the choice of experiments to be performed is made.

The results of an experiment with setting a on subsystem 1 are labelled by a real parameter s, taking values from a discrete set S_a of real numbers in the interval $[-1, 1]$. Equivalently, the result of an experiment on subsystem 2 is labelled by a parameter τ, taking values from a discrete set T_b in $[-1, 1]$. The sets of possible outcomes depend on the choice of experimental settings. The restriction of the outcome labels to the interval $[-1, 1]$ is a choice made only for convenience. It is essentially a measurement calibration, and it is important that the calibration be consistent across all experiments. The restriction to discrete outcome sets is not essential, but it is convenient for the proof of the Bell inequalities.

It is assumed that, for each pair of settings a, b, and every $z \in Z$, there is a probability function $p_{a,b}(s, \tau \mid z)$, which obviously sums to unity over all $s \in S_a$ and $\tau \in T_b$. These probabilities are used to define marginal probabilities:

$$p^1_{a,b}(s \mid z) \equiv \sum_{\tau \in T_b} p_{a,b}(s, \tau \mid z), \tag{3.1}$$

$$p^2_{a,b}(\tau \mid z) \equiv \sum_{s \in S_a} p_{a,b}(s, \tau \mid z), \tag{3.2}$$

Define $A_z(a, b)$ and $B_z(a, b)$ as the expectation values, for complete state z, of the outcomes of experiments on subsystems 1 and 2, respectively, when the settings are a, b.

$$A_z(a, b) \equiv \sum_{s \in S_a} s\, p^1_{a,b}(s \mid z) \tag{3.3}$$

$$B_z(a,b) \equiv \sum_{\tau \in T_b} \tau \, p^2_{a,b}(\tau \mid z) \tag{3.4}$$

Now define the expectation value of the product $s \cdot \tau$ of outcomes:

$$E_z(a,b) \equiv \sum_{s \in S_a,\, \tau \in T_b} (s \cdot \tau)\, p_{a,b}(s,\tau \mid z). \tag{3.5}$$

Bell-type inequalities follow from considerations of locality and causality that have been called *factorisability* or *Bell locality* (BL). This is formulated as the following factorisation condition on the response probability distribution of outcomes, conditional on the complete state z of the configuration.

(BL) For any a, b, z, there exist probability functions $p^1_a(s \mid z)$ and $p^2_b(\tau \mid z)$ such that

$$p_{a,b}(s,\tau \mid z) = p^1_a(s \mid z)\, p^2_b(\tau \mid z).$$

Note that, even though the probability factors into independent subsystem-1 and subsystem-2 probabilities conditional on the total system state z, the outcome in subsystem 1 does not depend on the setting of the experiment in subsystem 2, and vice versa. The condition (BL) is a consequence of the conjunction of two conditions, which have been called *parameter independence* and *outcome independence* (see Section 3.1, below).

If the factorisability condition (BL) is satisfied, then $A_z(a,b) = A_z(a)$ and $B_z(a,b) = B_z(b)$, so

$$E_z(a,b) = A_z(a) B_z(b). \tag{3.6}$$

The above definitions are valid for experiments with any number of discrete outcomes. The simplest case is that in which each experiment has only two distinct outcomes, labelled by ± 1.

To derive the inequalities, at least two separate experimental settings per subsystem must be considered, a, a', b, b'. Now consider the quantities

$$S_z(a, a', b, b') = |E_z(a, b) + E_z(a, b')| + |E_z(a', b) - E_z(a', b')|. \quad (3.7)$$

Now introduce a quantity related to $S_z(a, a', b, b')$, namely S_δ, where the expectation values in Equation 3.7 are averaged over δ, the distribution over states $z \in Z$.

$$\begin{aligned} S_\delta(a, a', b, b') = &|\langle E_z(a, b)\rangle_\delta + \langle E_z(a, b')\rangle_\delta| \\ &+|\langle E_z(a', b)\rangle_\delta - \langle E_z(a', b')\rangle_\delta|. \end{aligned} \quad (3.8)$$

Note that, in taking the expectation values of the quantities appearing on the right-hand side of this definition with respect to the same distribution δ, invoking the Measurement Independence Assumption. Under this assumption and the BL condition, the following is proved in J. F. Clauser et al. (1969).:

Bell's theorem — part 1: For $S_\delta(a, a', b, b')$ given by Equation 3.8, the following inequality holds:

$$S_\delta(a, a', b, b') \leq 2. \quad \text{(BT1)}$$

This is known as the *CHSH (Clauser–Horne–Shimony–Holt) inequality.*

Proof

From Equation 3.7

$$\langle S_z(a,a',b,b')\rangle_\delta = \langle |E_z(a,b)+E_z(a,b')|\rangle_\delta + \langle |E_z(a',b)-E_z(a',b')|\rangle_\delta.$$

Since the absolute value of the average of any random variable cannot be greater than the average of its absolute value, it is clear that

$$S_\delta(a,a',b,b') \leq \langle S_z(a,a',b,b')\rangle_\delta. \tag{3.9}$$

If BL is satisfied, then, using Equation 3.5 and the fact that $A_z(a)$ and $A_z(a')$ lie in the interval $[-1,1]$,

$$S_z(a,a',b,b') \tag{3.10}$$

$$= |A_z(a)(B_z(b)+B_z(b'))| + |A_z(a')(B_z(b)-B_z(b'))|$$

$$\leq |B_z(b)+B_z(b')| + |B_z(b)-B_z(b')|.$$

It is easy to check that

$$|B_z(b)+B_z(b')| + |B_z(b)-B_z(b')| = 2\max(|B_z(b)|,\, |B_z(b')|).$$

Since $B_z(b)$ and $B_z(b')$ also lie in the interval $[-1,1]$, from the previous step, for every z,

$$S_z(a,a',b,b') \leq 2. \tag{3.11}$$

Since this bound holds for every value of z, it must also hold for the expectation value of S_z.

$$\langle S_z(a, a', b, b')\rangle_\delta \leq 2. \tag{3.12}$$

This, together with Equation 3.9, yields the CHSH inequality and completes the proof of BT1. □

The final step of the proof of our Bell theorem is to show that a system described by a quantum mechanical state violates BT1, the CHSH inequality.

Bell's theorem — part 2: There is a quantum mechanical state and a set of experiments for which the CHSH inequality is violated.

$$\langle S_z(a, a', b, b')\rangle_\delta \not\leq 2. \tag{BT2}$$

The set of experiments used to test the CHSH inequality will be referred to as CHSH experiments. Here, consider one example involving polarisation-entangled photons, Figure 3.1.

Consider a pair of photons 1 and 2 propagating in the ζ-direction. Let ϕ_χ and ϕ_υ represent states in which photon j ($j = 1, 2$) is linearly polarised in the χ- and υ-directions, respectively. Neglecting the part of the state that describes movement in the ζ-direction, consider the following part of the state vector that describes the polarisation of the photons:

$$\Phi = \frac{1}{\sqrt{2}}\left(\phi_{\chi,1} \otimes \phi_{\chi,2} + \phi_{\upsilon,1} \otimes \phi_{\upsilon,2}\right), \tag{3.13}$$

which is invariant under rotation of the χ and υ axes in the plane perpendicular to ζ. The total quantum state of the pair of photons 1 and 2 is invariant under exchange of the two photons, as required by the fact that photons are integral-spin particles. Suppose now that photons 1 and 2 impinge respectively on the faces of birefringent crystal polarisation analyzers I and II, with the entrance face of each analyzer perpendicular to ζ. Each analyzer has the property of separating light

incident upon its face into two outgoing non-parallel rays, the *ordinary ray* and the *extraordinary ray*. The transmission axis of the analyzer is a direction with the property that a photon polarised along it will emerge in the ordinary ray (with certainty if the crystals are assumed to be ideal), while a photon polarised in a direction perpendicular to ζ and to the transmission axis will emerge in the extraordinary ray. See Figure 3.1 for a schematic of the experimental setup.

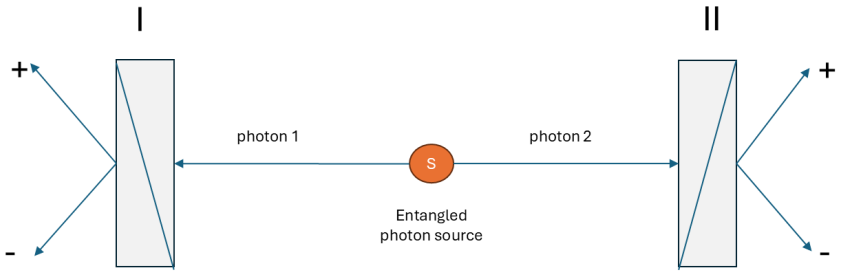


Figure 3.1.: Experimental Set-up

Photon pairs are emitted from the source, each pair quantum mechanically described by Φ of Equation 3.13. I and II are polarisation analyzers, with outcomes $s = 1$ and $\tau = 1$ designating emergence in the ordinary ray, while $s = -1$ and $\tau = -1$ designate emergence in the extraordinary ray. The crystals are also idealised by assuming that no incident photon is absorbed.

The expectation value, in state Φ, of the product of s and τ, is

$$E_\Phi(\mathbf{a}, \mathbf{b}) = \cos^2(\theta_{\mathbf{ab}}) - \sin^2(\theta_{\mathbf{ab}}) \tag{3.14}$$

$$= \cos(2\theta_{\mathbf{ab}}).$$

The details are calculated in Section 3.2, below.

Here $\theta_{\mathbf{ab}} = \alpha - \beta$ is the angle between the transmission axes of the two analyzers. The expectation value $E_\Phi(\mathbf{a}, \mathbf{b})$ is a correlation function, and it is a function of the angle between the transmission axes of the two analyzers. The correlation function is maximal when the transmission axes are aligned, and minimal when they are perpendicular.

An important insight of Bell's was that it is important to consider the less-than-perfect correlations obtained when the device axes are not aligned. Bell's theorem shows that no model satisfying the condition BL can reproduce the quantum correlations for all angles. This can be seen by considering the quantum predictions (Equation 3.13, Equation 3.14) and plugging them into the expression for S. For the case of polarisation-entangled photons

$$S_\Phi(\theta) = |2\cos(2\theta)| + |\cos(2\theta) - \cos(6\theta)|. \tag{3.15}$$

This takes on its maximum at $\theta = \pi/8$.

$$S_\Phi(\pi/8) = 2\sqrt{2} \approx 2.828. \tag{3.16}$$

3.1. Locality and causality assumptions

The key condition giving rise to the Bell inequality (BT1) is the factorisability condition (BL). This condition is motivated by considerations of locality and causality. It is also a purely mathematical relationship between probabilities. For the proof, it is only necessary to find one system described by a quantum state and a set of experiments for which the CHSH inequality is violated. However, for experimental tests

it is important to understand the physical assumptions that motivate the condition (BL) and ensure that these are met. In this section these assumptions are discussed.

3.1.1. Bell's Principle of Local Causality

Bell's Principle of Local Causality has evolved since the publication of Bell (1964). The details of this evolution are not important here, but rather the core idea is that the result of a measurement on one system be independent of the setting of a distant measurement, provided the two measurements were not coordinated in their joint past.

In his final contribution (Bell and Aspect (2004)), Bell derives the factorisability condition (BL) from the *Principle of Local Causality*. He bases his analysis on the following:

- relativity prohibits superluminal causation;
- the cause-effect relation is asymmetric in time, with causes temporally preceding their effects;
- in a relativistic spacetime, events at spacelike separation are taken to have no temporal order.

Bell's final statement of the Principle of Local Causality is as follows (see Figure 3.2):

> **Principle of Local Causality** (PLC): A theory will be said to be locally causal if the probabilities attached to values of local beables in a spacetime region 1 are unaltered by specification of values of local beables in a spacelike separated region 2, when what happens in the backward light cone of 1 is already sufficiently specified, for example by a full specification of local beables in a spacetime region 3.

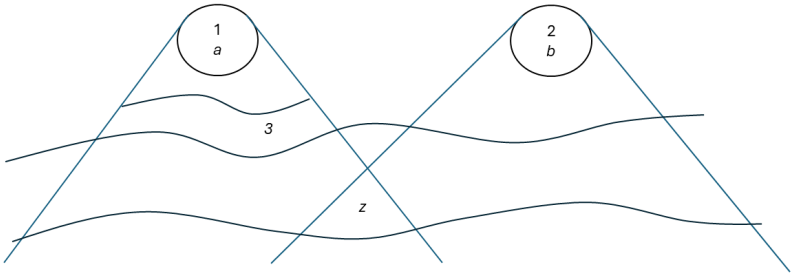


Figure 3.2.: Principle of Local Causality

Bell's term *beable* means any element of a physical theory that is taken to correspond to something physically real; therefore a beable is an entity in the Real Sphere as described in Critical ontology and layers of reality, and *local beables* pertaining to a spacetime region are those contained within that region. The term was, of course, introduced to generalise beyond the term *observable* in textbook QM.

Note that the preparation state z is a complete specification of the properties of the entangled system at the moment of generation, and hence is in the backward light cone of both region 1 and region 2.

The formulation of the local causality condition is motivated by Bell (1976) with the remark,

> Now my intuitive notion of local causality is that events in **spacetime region 2** should not be "causes" of events in **spacelike separated region 1**, and vice versa. But this does not mean that the two sets of events should be

> uncorrelated, for they could have common causes in the overlap of their backward light cones.

The system state z is a common cause of correlations between the two subsystems. Correlations between two variables that are not in a cause-effect relation to each other can be explained by some common cause.

Bell then proceeds to formulate the condition of local causality, which is that a full specification of beables in the overlap of the backward light cones of the spacetime regions 1 and 2 should screen off correlations between them. Implicit in this is that correlations between two variables be susceptible to causal explanation, either via a causal connection between the variables, or via a common cause.

The Principle of Local Causality follows from the conjunction of three assumptions, all of which were explicitly formulated by Bell:

- (Temporal Asymmetry of Causality) Any cause of an event lies in its temporal past, and not in its temporal future.
- (Lorentz Invariance) The relation of temporal precedence is invariant under Lorentz boosts.
- Common cause.

3.1.2. Parameter independence and outcome independence

The condition BL of factorisability is the application, to the particular setup of CHSH experiments, of Bell's Principle of Local Causality. As shown, it can be thought of as the conjunction of the condition of causal locality and the common-cause principle. In this subsection those conditions are applied to the setup of CHSH experiments.

On the assumption that the experimental settings can be treated as free variables, whose choice of setting on one wing is made at spacelike separation from the experiment on the other, a dependence of the probability of the outcome of one experiment on the setting of the

other would seem straightforwardly to be an instance of a nonlocal causal influence. The condition that this not occur can be formulated as follows.

(PI) For all parameter settings a, a', b, b', all z, and all $s \in S_a$ and $\tau \in T_b$,

$$p^1_{a,b}(s \mid z) = p^1_{a,b'}(s \mid z); \tag{3.17}$$

$$p^2_{a,b}(\tau \mid z) = p^2_{a',b}(\tau \mid z). \tag{3.18}$$

This is the condition that has come to be known as *parameter independence*, following Shimony (1986), Shimony (1990)).

For fixed values of the experimental settings, Bell's Principle of Local Causality entails that the outcomes of the experiments on the two systems be independent, conditional on the specification z of the complete state of the system at the source. This is the *outcome independence* condition.

(OI) For each pair of parameter settings a, b, and all z, and all $s \in S_a$ and $\tau \in T_b$,

$$p_{a,b}(s, \tau \mid z) = p^1_{a,b}(s \mid z)\, p^2_{a,b}(\tau \mid z). \tag{3.19}$$

A violation of PI would seem to permit superluminal signalling, but this requires an additional assumption: that the state of the system be controllable. It would not hold for a theory on which there are principled limitations on the ability of would-be signallers to control the states of the systems they are dealing with; an example of a theory of this sort is Bohmian mechanics, which violates PI but does not permit superluminal signalling. In some of the literature, PI has incorrectly been treated as equivalent to no-signalling.

3.1.3. Explicit experimental assumptions

3.1.3.1. Unique experimental outcomes

Of the potential outcomes of a given experiment, one and only one occurs, and hence it makes sense to speak of *the* outcome of an experiment. However, Everettian, or "many-worlds", interpretations state that all potential outcomes actually occur. Even in these interpretations, though, each individual world has a unique outcome.

3.1.3.2. Experimental settings as free parameters

It is assumed that the complete state z is always sampled from the same probability distribution δ, no matter what choice of experiment is made, and that, for this reason, the subset of experiments corresponding to any given choice of settings is a fair sample of the distribution of z. This will be discussed in more detail in Bell's theorem and metaphysics.

3.1.3.3. Timing of experimental outcomes

In experiments, care should be taken that, for the two subsystems, the choice of experimental setting and the registration of results take place at spacelike separation. It is assumed that experiments have unique results. The question arises as to *when* the unique result emerges. It is typically assumed that the result is definite once a detector has been triggered or the result has been recorded in a computer memory. This difficulty will remain unresolved until the quantum mechanical measurement problem is solved.

3.2. Appendix: quantum mechanical calculation of the CHSH inequality

The joint measurement probabilities of two entangled photons are calculated, their correlation function is found, and it is shown how they violate the limits of classical physics.

3.2.1. 1. The initial entangled state

In an optical setup using spontaneous parametric down-conversion (SPDC), a pair of photons are generated and sent to measurement equipment A and B in a maximally entangled polarisation state described by the symmetric Bell state, Equation 3.13-:

$$\Phi = \frac{1}{\sqrt{2}} \left(\phi_{\chi,1} \otimes \phi_{\chi,2} + \phi_{\upsilon,1} \otimes \phi_{\upsilon,2} \right),$$

Here, ϕ_χ and ϕ_υ represent horizontal and vertical linear polarisations. This state means that if settings a at A and b at B both measure in the χ or υ direction, they will always get identical results (either both χ or both υ).

3.2.2. 2. The measurement bases

Settings a and b can be independently rotated.

- the a setting of polariser I is the angle α;
- the b setting of polariser II is the angle β.

A polariser at angle θ measures whether a photon is polarised along that angle ($+1$ result) or perpendicular to it (-1 result). Write these measurement states as:

$$\psi_{+\theta} = \cos\theta\phi_\chi + \sin\theta\phi_\upsilon$$

$$\psi_{-\theta} = -\sin\theta\phi_\chi + \cos\theta\phi_\upsilon$$

3.2.3. 3. Calculating joint probabilities

Require the probability that the settings a and b both detect a photon passing through their polarisers $(+1, +1)$. According to Born's rule, this is the squared magnitude of the inner product of the measurement state and the initial state:

$$p_{++}(\alpha, \beta) = |(\psi_{+\alpha} \otimes \psi_{+\beta}, \Phi)|^2$$

First, let us find the inner product:

$$(\psi_{+\alpha} \otimes \psi_{+\beta}, \Phi)$$

$$= \frac{1}{\sqrt{2}}\left(\left(\phi_{\chi,1}\cos\alpha + \phi_{\upsilon,1}\sin\alpha\right) \otimes \left(\phi_{\chi,2}\cos\beta + \phi_{\upsilon,2}\sin\beta\right), \left(\phi_{\chi,1} \otimes \phi_{\chi,2} + \phi_{\upsilon,}\right.\right.$$

Because $(\phi_\chi, \phi_\chi) = 1$ and $(\phi_\chi, \phi_\upsilon) = 0$, this simplifies to:

$$\left(\psi_{+\alpha} \otimes \psi_{+\beta}, \Phi\right) = \frac{1}{\sqrt{2}}(\cos\alpha\cos\beta + \sin\alpha\sin\beta)$$

Using the trigonometric identity $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$,

$$\left(\psi_{+\alpha} \otimes \psi_{+\beta}, \Phi\right) = \frac{1}{\sqrt{2}}\cos(\alpha - \beta)$$

Squaring this gives the joint probability of both getting $+1$:

$$p_{++}(\alpha,\beta) = \frac{1}{2}\cos^2(\alpha-\beta)$$

By doing the same for the other combinations (for example, using $\phi_{-\alpha}$ or $\phi_{-\beta}$), the full set of joint probabilities is:

- p_{++} (both pass): $\frac{1}{2}\cos^2(\alpha-\beta)$
- p_{--} (both blocked): $\frac{1}{2}\cos^2(\alpha-\beta)$
- p_{+-} (a passes, b blocked): $\frac{1}{2}\sin^2(\alpha-\beta)$
- p_{-+} (a blocked, b passes): $\frac{1}{2}\sin^2(\alpha-\beta)$

3.2.4. 4. The expectation value (correlation function)

The correlation function $E(\alpha,\beta)$ is the average product of their measurement outcomes. Weighting the matching results ($+1\times+1=1$ and $-1\times-1=1$) positively, and the opposing results ($+1\times-1=-1$) negatively:

$$E(\alpha,\beta) = p_{++} + p_{--} - p_{+-} - p_{-+}$$

Substituting the probabilities just calculated:

$$E(\alpha,\beta) = \left(\frac{1}{2}\cos^2(\alpha-\beta) + \frac{1}{2}\cos^2(\alpha-\beta)\right) - \left(\frac{1}{2}\sin^2(\alpha-\beta) + \frac{1}{2}\sin^2(\alpha-\beta)\right)$$

$$E(\alpha,\beta) = \cos^2(\alpha-\beta) - \sin^2(\alpha-\beta)$$

Using the identity $\cos(2\theta) = \cos^2\theta - \sin^2\theta$, the quantum correlation function is derived as:

$$E(\alpha,\beta) = \cos(2(\alpha-\beta))$$

This is exactly the result as stated in Equation 3.14, with $\alpha - \beta = \theta_{ab}$, confirming that the quantum mechanical predictions for the correlation function violate the CHSH inequality for certain choices of angles α and β.

Part III.

A transactional quantum theory

4. Possibilist Transactional Quantum Theory (PTQT)

Ruth Kastner's "Relativistic (Possibilist) Transactional Interpretation" (PTI) gives a very explicit and technically different answer to Bell locality than Cramer's original Transactional Interpretation (TI). Her answer is grounded in *direct-action quantum field theory (DAQFT)* and an ontology of *real physical possibilities* leading to spacetime events.

This chapter presents a concise conceptual summary of Possibilist Transactional Quantum Theory (PTQT), focusing on two questions:

1. How PTQT can be reformulated within the probabilistic reconstruction of quantum theory (Chapter 1)
2. How PTQT can be used to describe the ontological status of the quantum substratum.
3. How reformulated PTQT addresses Bell locality and correlations without invoking spacetime superluminal causation.

To address the first point, a simple change will be made. All talk of waves will be eliminated. The immediate changes are:

- Offer waves (OWs) will be replaced by *emission states* (ES) of the emitter.
- Confirmation waves (CWs) will be replaced by *absorber states* (AS)

4.1. The quantum substratum

Kastner's PTI is often described as an interpretation of quantum mechanics (Kastner (2022)). In this chapter, it is treated as a physical theory with explicit ontological commitments; for that reason, the name *Possibilist Transactional Quantum Theory* (PTQT) is used to reflect this status.

The core claim is that a *quantum substratum* of physically real possibilities underlies, and specifically causes, observable spacetime events. The theory addresses both the measurement problem and Bell-type correlations. Except for the ad hoc measurement step in standard quantum theory all quantum processes take place in the substratum. They are not spacetime processes. In PTQT, the substratum is a domain of potentiality rather than a collection of actual spacetime objects. It is not directly observable; its structure is inferred from the formalism and from the pattern of actualszed events.

A potentiality-based reading of quantum theory has precedents (e.g., Heisenberg (1958)). PTQT extends this by tying potentiality to a transactional mechanism that yields concrete events. The substratum is an aspect of the inorganic layer in the Real Sphere. It is an extension of the more classical conception that is to be found in Hartmann's writings. Quantum unitary dynamics is a process in the substratum, and the non-unitary actualisation of events is a process that creates events in spacetime. The spacetime events are also items in the inorganics and exist in the Real Sphere. The substrate exists as the domain of real possibilities and the spacetime is the domain of real events.

On this view, spacetime is not fundamental. Events may be ontologically prior, with spacetime emerging from relations among events. This reframes matter-like entities as emergent from an event-structured process rather than as fundamental substances.

PTQT describes actualisation as a stochastic selection among physically possible transactions:

1. A relativistic quantum state evolves into a state supporting multiple possible transactions.
2. The context defines transition amplitudes/probabilities over distinct possibilities.
3. One possible transaction is actualised through stochastic selection.
4. The outcome is a spacetime event and an updated set of future possibilities.

This actualisation process is non-unitary and stochastic and it is a process that creates events in spacetime. Measurement is modeled as selection among multiple incipient transactions, sometimes described as spontaneous symmetry breaking in the transaction set. The resulting event is an actualised outcome in spacetime, while unrealised alternatives remain non-actualised possibilities.

Measurement actualisation is spontaneous symmetry breaking In PTQT:

- Multiple "incipient transactions" exist as possibilities.
- A single transaction is actualised via *spontaneous symmetry breaking* according to Kastner but this is an analogical description. There is no Higgs mechanism in the substratum. The selection is stochastic and non-unitary. (the discussion of the role of virtual photons, Chapter 5 will help clarify the mechanism of selection).

This actualisation creates spacetime events.

4.2. Time in PTQT

If time in the empirical sense is emergent then what is it that indexes the dynamics in the pre-spacetime quantum substrate? In the PTQT spacetime is not a fundamental container. Instead, physical spacetime and its temporal structure emerge from a deeper, extra-spatiotemporal realm called the quantum substratum (Kastner (2022)). The time mechanism in this framework operates through the following core concepts:

- In PTQT, the quantum substratum consists of quantum states existing as potentiae (possibilities or unactualised offers and responses). Time is a process index in this substratum and spacetime events and their structural connections are established exclusively by "actualised transactions". These transactions take place when:
  - A quantum emitter sends an offer state and an absorber responds, a transaction can become actualised (a process corresponding to objective wave function collapse).
  - An actualised transaction instantiates a discrete emission event, an absorption event, and the invariant spacetime interval connecting them. Spacetime effectively "crystallises" out of these events.
- PTQT rejects the "block universe" ontology where past, present, and future all exist simultaneously.
  - In the quantum substratum, the primary temporal feature is change driven by the state transitions of interacting quantum systems.
  - The "Now" is the shifting domain of active actualisation. The future does not exist yet because it consists only of unactualised possibilities within the substratum. - A time index (t) is merely a way of indexing and mapping these

changes after they have been actualised into the emergent fabric of spacetime.

- Standard quantum mechanics operates on time-symmetric laws, making it a mystery why time only moves forward. PTQT solves this by introducing a physical mechanism for temporal symmetry breaking:

  - The unitary dynamics of unactualised possibilities in the substratum is fundamentally time-symmetric.
  - However, the non-unitary phase where an absorber responds and a virtual photon exchanges transforms the time-symmetric propagator into the time-asymmetric Feynman propagator (which propagates positive energy forward in time).
  - This actualisation process inherently breaks time symmetry, generating a unidirectional arrow of time and grounding the Second Law of Thermodynamics.

In summary, normal time is relational and emergent: the quantum substratum itself is pre-spatiotemporal, while the linear flow of time perceived is an empirical byproduct of discrete, irreversible quantum transactions breaking symmetry at the micro-level.

As the quantum substratum is pre-spatiotemporal, how can we speak of "unitary evolution" (like the solution of the Heisenberg equation or quantum operators) which explicitly depends on a time parameter? Ruth Kastner resolves this by introducing a strict distinction between two fundamentally different types of time: Coordinate Time (which is emergent and relational) and Process Time (which is fundamental and non-spatial). Here is how unitary dynamics can function in a substratum without relying on physical spacetime:

4.2.1. Process Time vs. Coordinate Time

In PTQT, the time parameter (t) found in standard quantum mechanical equations does not refer to the ticking of a clock in physical space. Instead, it refers to Process Time:

- Process Time: This is a sequential, logical ordering of changes in quantum states. It represents the generation of possibilities—offers being sent and responses being generated. It is a time of becoming, not a time of extension.
- Coordinate Time: This is the time we measure with macro-clocks (e.g., x^0 in relativity). It only exists after a transaction has collapsed the quantum state and created a physical event.

Therefore, unitary dynamics describes the evolution of possibilities in Process Time before they ever "solidify" into the Coordinate Time of our physical universe. Because the substratum lacks spatial distance, all quantum systems within it are fundamentally in a state of immediate, non-local availability to one another.Therefore: - Unitary evolution in this realm is not a movement through space over time. The time t in the unitary operator $U(t) = e^{-iHt}$ acts as a relational index for these internal state changes, rather than a measure of duration through an external physical container.

4.3. Bell locality in PTQT

Bell locality (Chapter 3) applies only to statistics of spacetime events. Kastner argues that Bell's locality condition applies only to actualied spacetime events in PTQT but are not relevant to the pre-spacetime quantum substratum where emission states and absorber states exist.

In PTQT, the states involved in quantum processes are not spacetime objects; they are *possibilities described by state space*. Thus, Bell's locality assumption does not apply at the level where entanglement

correlations originate. That is in the quantum substratum. The Implication is that Bell's theorem does not constrain PTQT because the correlations arise *before* spacetime events exist — in a domain where "Bell locality" is not defined.

PTQT's response to Bell's theorem is that Bell locality applies to *actualised spacetime events*, not to the pre-spacetime domain in which transactional possibilities are defined.

Emission states and absorber states are treated as elements of that possibilist structure, not as classical signals propagating through spacetime. Under this interpretation, Bell-violating correlations are generated by nonlocal constraints at the possibilist level, rather than by superluminal causal influence between detectors.

PTQT therefore accepts nonlocal correlations while aiming to preserve relativistic covariance: the key process is modeled as *pre-spatiotemporal*, avoiding a preferred foliation.

Kastner explicitly accepts *nonlocal correlations*, but denies that they imply *spacetime nonlocal causation*. The causal aspect belongs to the substratum. In PTQT, the transaction between emitter and absorber is a *pre-spacetime process* realised through the exchange of a virtual photon. It is not a signal traveling through spacetime, so it does not violate relativistic locality. The implication is that Bell violations arise from a *global constraint* in the quantum substratum, not from superluminal influences.

4.4. PTQT correlations

PTQT uses direct-action theory (DAQFT) to explain correlations and events without hidden variables. It is built on Davies' quantum direct-action theory, where interactions occur through:

- Emission states (ES) - actually the quantum state of the emitter (ES) [1]
- Confirmation waves (CW) - actually the quantum state of the absorber (AS)

These are not fields propagating locally in spacetime but *nonlocal interaction amplitudes*. The coupling constant (charge) is interpreted as the amplitude for generating ES/AS.

Bell's factorisability condition fails because the underlying ontology is *nonlocal by construction*, but not in a way that threatens relativity.

4.5. Retrocausality and atemporal constraint language

A recurring interpretive issue is whether PTQT should be described as retrocausal. A useful distinction is between:

- Dynamical retrocausality (literal backward-in-time causal propagation), and
- Atemporal/all-at-once constraint structure (global consistency conditions without propagating backward signals).

For PTQT, the second description is typically more precise: advanced mathematical components need not imply dynamical backward causation.

[1] The offer wave is the quantum state of the emitter, which is a physically real possibility in the substratum. It is not a spacetime field or signal. Kastner often uses the term "offer wave" to describe this possibility amplitude, but it is important to note that it is not a propagating field in spacetime. It is a nonlocal amplitude in the pre-spacetime substratum. Similarly for the confirmation wave, which is the quantum state of the absorber (AS).

4.6. Haag locality clash with PTQT

In axiomatic foundation for quantum theory as proposed in Haag (1996), Haag locality is a spacetime concept, and PTQT denies that spacetime is fundamental. In PTQT the operators act on pre-spacetime quantum states Therefore, Haag locality cannot be a fundamental axiom of quantum theory if PTQT is correct[2].

4.6.1. Haag locality presupposes spacetime

Haag's Local Quantum Physics (LQP) is built on three pillars:

- Local operator algebras assigned to *spacetime regions*
- Microcausality: operators in spacelike-separated regions commute
- Haag duality: the algebra of a region equals the commutant of its causal complement

All of these pillars require:

- A background spacetime
- A causal structure
- Localisable observables
- Region-indexed operator algebras

So Haag locality is intrinsically spacetime-dependent. It invokes nothing more fundamental than spacetime geometry. It is an *axiomatisation of QFT on spacetime*.

[2]Note that the point here is the inconsistency between the two theories

4.6.2. PTQT denies spacetime at the fundamental level

In Kastner's PTQT emitter states (ES) and absorber states (AS) exist in a pre-spacetime substratum

- Incipient transactions are *pre-spacetime* possibilities. There is a substratum time but this a time intrinsic to the non-zero rest mass particles concerned in the transaction.
- The combination of an ES with potential absorber states defines a set of possible transactions.
- The combination of the ES and each AS defines a set of transition probabilities for the possible transactions.
- An event is actualised when one of the possible transactions is selected via a stochastic process. This is a spontaneous process that selects one transaction from the set of possibilities.
- Actualisation *creates* a spacetime interval event.

- Spacetime is not fundamental, but emergent from transactions
- The emergent space-time is an attribute of the inorganic layer in a general ontology.

Therefore, Before actualisation, there are no spacetime regions, no causal complements, and no operator algebras indexed by regions. Haag locality simply has no domain of definition at the fundamental level. Therefore Haag locality cannot be fundamental. If spacetime is emergent then Haag locality is also emergent. It is a constraint on actualised events, not a fundamental axiom of the quantum substratum.

Thus Haag locality is at best an emergent constraint on actualised events, not a fundamental axiom of the quantum world. This means Haag's framework cannot be the foundational axiomatic structure of quantum theory if PTQT is correct. It becomes a derived effective theory that is valid only after actualisation, not before. It is a phenomenological constraint on the emergent spacetime layer, not a fundamental axiom of the quantum substratum.

4.6.3. The consequence for axiomatic foundations

If PTQT is correct:

- Haag locality cannot be a fundamental axiom
- Haag–Kastler algebraic QFT cannot be the foundational ontology
- Local operator algebras are emergent structures
- The axioms of LQP describe only the emergent spacetime layer

This is analogous to:

- Thermodynamics not being fundamental if statistical mechanics is correct
- Continuum elasticity not being fundamental if atomic physics is correct

Haag locality becomes a phenomenological constraint on the emergent spacetime layer.

4.6.4. What an operator algebra *is* in Haag's framework

In Haag's Local Quantum Physics (LQP), the basic objects are:

- C*-algebras or von Neumann algebras
- indexed by spacetime regions $\mathcal{A}(\mathcal{O})$
- satisfying locality:

$$[\mathcal{A}(\mathcal{O}_1), \mathcal{A}(\mathcal{O}_2)] = 0 \quad \text{if } \mathcal{O}_1 \text{ and } \mathcal{O}_2 \text{ are spacelike separated}$$

Everything — locality, duality, isotony, covariance — depends on spacetime geometry.

Without spacetime, the indexing collapses and the axioms cannot even be *stated*.

4.6.5. PTQT's quantum substratum has *no spacetime*

In Kastner's PTQT:

- Offer waves (OWs) and confirmation waves (CWs) exist in a pre-spacetime substratum
- Incipient transactions are *atemporal* possibilities
- Actualisation (NU-process) *creates* a spacetime interval
- Before actualisation, there are no regions, no causal complements, no metric, no light cones.

Thus:

> There is no structure on which a Haag–Kastler net of algebras could be defined.

The substratum is not a manifold, not a causal set, not a metric space, not a topological space.
It is a Hilbert-space possibility structure, not a spacetime.

One might try to salvage locality by saying:

> "Let's index algebras by something other than spacetime — e.g., degrees of freedom, emitters, absorbers, or ES/AS channels."

But this fails because the notion of locality exists in the substratum as locality is a relation between *regions* and the substratum has no regions. Therefore operator locality is undefined in the substratum. Not violated — simply meaningless.

4.6.5.1. What *can* exist in the substratum

PTQT *does* have algebraic structure, but it is:

- global, not local
- constraint-based, not region-based
- possibility-based, not event-based
- atemporal, not temporal
- nonlocal in Bell's sense, but non-signalling

You can have:

- Hilbert space
- operators representing OWs and CWs
- amplitudes
- global constraints on entangled systems
- Born-rule weights for incipient transactions

But you cannot have:

- local operator algebras

- nets of algebras
- microcausality
- Haag duality
- isotony
- region-indexed structures

Because all of these require spacetime, which PTQT says is not fundamental. If PTQT is correct, Haag locality is not fundamental and cannot serve as an axiomatic foundation for quantum theory. Haag locality becomes:

- an emergent constraint
- applicable only after actualisation
- valid only within spacetime
- not part of the fundamental ontology

4.6.5.2. Operators exist in the substratum — but they are *not local operators*

PTQT's quantum substratum is:

- pre-spacetime
- atemporal
- nonlocal in Bell's sense
- a Hilbert-space possibility structure

In this domain, there are:

- Hilbert space
- linear operators
- self-adjoint operators
- projectors
- unitary operators
- tensor-product structure
- global constraints on entangled states

These operators form a perfectly good algebraic structure: a C*-algebra or von Neumann algebra of operators on Hilbert space. This is the same algebraic backbone used in the probabilistic reconstruction of quantum mechanics (Chapter 1). PTQT supports a global operator algebra, not a local one. It is a Hilbert-space operator algebra and the operators act on:

- Emitter states
- Absorber response states
- entangled states
- prepared states
- global possibility structures

This is the usual algebra of quantum theory:

$$\mathcal{B}(\mathcal{H})$$

or a suitable C*-subalgebra. It supports unitary evolution - the U-process is governed by:

$$U(t) = e^{-iHt}$$

where t is Process Time (internal periodicity), not spacetime time.

- It supports projectors
  - Incipient transactions correspond to projectors onto matched OW–CW subspaces.
- It supports Born-rule weights
  - The algebra supports the spectral decomposition needed for probabilities.
- It supports global constraints
  - Entangled systems impose nonlocal constraints on the operator algebra.
- It does *not* support locality
  - Because locality requires spacetime, which is emergent.

Once a transaction actualises:

- a spacetime interval is created
- emitter and absorber actualise spacetime events

At this stage quantum mechanics has done its job. There is no need for operators to be assigned to regions. Thus, local operator algebras are not even emergent structures let alone fundamental ones.

Operators in the PTQT substratum must have an algebraic structure — and they do.
But it is a global Hilbert-space operator algebra, not a Haag–Kastler local algebra.

Therefore:

- Global operator algebra: fundamental
- Local operator algebra: not applicable
- Haag locality: not applicable
- PTQT substratum: algebraic but nonlocal
- Spacetime locality: created by actualisation

This is exactly the distinction needed to reconcile PTQT with operator theory while rejecting Haag locality as a foundational axiom.

4.7. State preparation, spacetime and the quantum substratum

State preparation has similarities with measurement. A measurement leaves the system under investigation in a certain state. State preparation in Kastner's Relativistic Transactional Interpretation (PTQT) (Kastner (2022)) influences the quantum substratum by *selecting and shaping the set of incipient transactions* that exist as physically real possibilities prior to spacetime actualisation. It does so through isolating the system under investigation and then controlling its interactions with a preparatory absorber.

The setting up of the preparatory absorber takes place in spacetime, but the state preparation is not in spacetime; it is a constraint on the *pre-spacetime possibility structure* from which actualised events emerge.

In PTQT, state preparation modifies the quantum substratum by altering the configuration of emission states of the system under investigation and absorber states that exist as pre-spacetime possibilities. These prepared states determine *which incipient transactions are available* and therefore *which spacetime events can emerge* through the non-unitary (NU) actualisation process.

Kastner describes the quantum substratum as a pre-spacetime domain consisting of quantum systems with rest mass and internal periodicities that act as "internal clocks" and define inertial frames. The substratum contains in Kastner's terminology:

- Offer waves (OWs) generated by emitters
- Confirmation waves (CWs) generated by absorbers
- Incipient transactions: matched OW–CW pairs representing *possible* energy-transferring events

These are representations of possibilities in Hilbert space.

The generalised probability reconstructions of quantum mechanics and now the enhanced PTQT show that the quantum substratum is a pre-spacetime domain in which the possibilities for actualisation are constrained by the global structure of the system. State preparation is a process that constrains the substratum and thereby influences which spacetime events can emerge. There is no place other than historical respect for Schrodinger to talk about waves. So instead of describing the processes in terms of *offer waves* and *confirmation waves*, we can describe them in terms of the *emission states* (ES) of the system under investigation and the *absorber states* (AS).

State preparation—whether by measurement, filtering, or controlled interactions—changes the quantum state that undergoes unitary evolution (the U-interaction).

Prepared states determine:

- which ES exist,
- their amplitudes,
- their phase relations,
- the set of potential absorbers,
- and therefore the structure of incipient transactions.

Thus, state preparation reconfigures the landscape of possibilities in the substratum.

4.7.1. How state preparation influences the substratum

It changes the set of emission states by preparing a quantum state, which changes the ES of a system. Since ES are the "invitations to transact," modifying them directly alters the substratum's possibility structure. It changes which absorber states can respond to the emission states, and therefore which incipient transactions can form. Absorbers respond only to ES components compatible with their coupling.

State preparation can: - suppress certain components (e.g., filtering out momentum modes), - enhance others, - or entangle systems so that absorber states reflect joint absorber behaviour.

It changes the pattern of incipient transactions by forming them wherever ES–AS matches occur. State preparation determines:

- which matches are possible,
- their relative weights,
- their global constraints (especially in entangled systems).

This is crucial because actualisation selects one incipient transaction via what Kastner calls spontaneous symmetry breaking. There is no explanatory mechanism for this "symmetry breaking" in PTQT, but the selection of the actual outcome is more analogous to random sampling. But there is no sampling mechanism. All there is, is a dispositional weighted set of tendencies for each possibletransaction and a stochastically selected outcome.

4.7.2. Why this matters for measurement and Bell correlations

Because state preparation determines the set of incipient transactions, it also determines:

- the correlations that can appear,
- the global constraints on entangled systems,
- and the structure of possible actualisations.

This is why PTQT can explain Bell correlations without spacetime nonlocality:
the correlations arise from global constraints in the substratum, not from signals between spacetime events.

State preparation influences the quantum substratum in PTQT by altering the configuration of emission states through selection of a preferred absorber state via transactions in the pre-spacetime substratum.

4.8. Conclusion

PTQT has a pre-spacetime, possibilist ontology in which:

1. Quantum possibilities are physically meaningful
2. Spacetime events are actualised transaction outcomes,
3. Bell correlations arise from nonlocal possibilist constraints outside spacetime, and
4. Relativistic structure is preserved by avoiding spacetime superluminal signaling.

4.9. Photons in the quantum substratum spacetime

A *real photon* in QFT is an on-shell excitation of the electromagnetic field that can be detected, while a *virtual photon* is an off-shell internal line in a perturbative calculation that never appears as an observable particle. That is the essential distinction.

A *real photon* satisfies the physical dispersion relation $E^2 = p^2c^2$ and can be emitted, absorbed, or counted. A *virtual photon* does not satisfy this relation; it is a mathematical intermediate in the perturbation expansion of the S-matrix and cannot be directly observed. Physics Forums

Real photons (on-shell) are:

- Actual quanta of the electromagnetic field.
- Appear in the *external legs* of Feynman diagrams.
- Are **detectable**: you can measure their energy, momentum, polarisation.

Their key features are

- *On-shell:* obey $E^2 = p^2c^2$.
- *Long-lived:* propagate freely.
- *Transfer energy–momentum:* they mediate real radiation.
- *Create or annihilate excitations* in detectors or atoms.

Real photons correspond to actual events — "something happened" in the experimental sense. Springer

Virtual photons (off-shell) are

- Internal lines in Feynman diagrams.
- Mathematical devices in the Dyson expansion of the S-operator.
- Never appear as asymptotic states.

Their key features are:

- *Off-shell:* do *not* satisfy $E^2 = p^2c^2$.
- *Short-lived:* exist only within the integral over intermediate momenta.
- *Not observable:* cannot be detected or counted.
- *Encode interactions:* they represent how charged particles influence each other.

Virtual photons are not just part of the *dressing* of charged particles, where in standard quantum field theory they appear as aspects of the calculation. In PTQT they make the transaction.

Feature	Real photon	Virtual photon
On/off shell	On-shell	Off-shell
Appears where?	External legs	Internal lines
Detectable?	Yes	No
Energy–momentum relation	Satisfies (E^2 = p^2 c^2)	Violates it
Role	Physical radiation	Mediates interactions mathematically
Lifetime	Arbitrary	Constrained by propagator structure

Feature	Real photon	Virtual photon
Physical status	Actual field excitation	Bookkeeping device in perturbation theory

Why the distinction matters

- *Only real photons* correspond to measurable radiation.
- *Virtual photons* are indispensable in calculations but their physical existence is in the quantum substratum.
- Confusing the two leads to misconceptions like "vacuum is full of particles popping in and out of existence." In QFT, virtual particles are *terms in an integral*, not literal transient objects, but in PTQT they are essential for completing the transaction.

A photon in the substratum completes the transaction between the ES and the set of possible AS's.

Ontological status of the virtual photon in the quantum substratum (PTQT):

- It is not a particle and not a spacetime object.

- It completes the transaction between the emitter and the selected absorber.
- It exists only in the quantum substratum, which is *not* spacetime and has no metric, no geometry, no trajectories.

Ontological statusof the real photon in spacetime (PTQT):

- It is the actualised transfer of conserved quantities (energy–momentum).

- It is a spacetime connection between an emission event and an absorption event.

- It is a causal-set link in the emergent spacetime structure.
- Key properties:
  - Exists only after a transaction (NU-process).
  - Defines spacetime events — absorption events become actual spacetime nodes.
  - Carries energy–momentum — unlike the transaction photon.
  - Has a spacetime trajectory (null geodesic) *only after* spacetime exists.

Feature	Photon in quantum substratum	Photon in spacetime
Ontological status	Possibility (ES)	Actualised energy transfer
Domain	Pre-spacetime substratum	Emergent spacetime
Trajectory	None	Null geodesic between events
Time structure	Time-symmetric proto-time	Time-asymmetric spacetime time
Role	Defines potential interactions	Creates spacetime structure

5. Appendix: Photons in the quantum substratum and spacetime

In PTQT the difference between real and virtual photons is explicitly defined by whether a complete, energy-conserving transaction is formed. Instead of treating both as different "types" of independent field excitations, PTQT views them as fundamentally distinct stages of relational interactions between charged particles.

##Real photons

A *real photon* in QFT is an on-shell excitation of the electromagnetic field that can be detected, while a *virtual photon* is an off-shell internal line in a perturbative calculation that never appears as an observable particle. A *real photon* satisfies the physical dispersion relation $E^2 = p^2c^2$ and can be emitted, absorbed, or counted. A *virtual photon* does not satisfy this relation; it is a mathematical intermediate in the perturbation expansion of the S-matrix and cannot be directly observed.

Real photons (on-shell) appear in the *external legs* of Feynman diagrams and are *detectable*: you can measure their energy, momentum, polarization. Their key features are

- *On-shell:* obey $E^2 = p^2c^2$.
- *Long-lived:* propagate freely.

- *Transfer energy–momentum:* they mediate real radiation.
- *Create or annihilate excitations* in detectors or atoms.

Real photons correspond to actual events — "something happened" in the experimental sense.

In PTQT a photon becomes "real" when an excited emitter successfully completes a "handshake" with a specific absorber rather than just a potential one.

- The emitter state and the absorber states set up a set of possibilities.
- If and only if the conservation laws are perfectly satisfied between those two specific matter states, a non-unitary collapse occurs, which is a stochastic transition from the possible to the actual. The energy is transferred, and a real photon is actualised.
- While standard quantum physics defines a real photon as something that can theoretically travel infinitely through space without an absorber, in PTQT a real photon always has an absorber on the other end. It is bounded by its emission and absorption events.

5.1. Virtual photons

Virtual photons in QFT are internal lines in Feynman diagrams. They can be considered to be mathematical devices in the Dyson expansion of the S-operator.

Their key features are:

- *Off-shell:* do *not* satisfy $E^2 = p^2c^2$.

- *Short-lived:* exist only within the integral over intermediate momenta.

- *Not observable:* cannot be detected or counted.

- *Encode interactions:* they represent how charged particles influence each other.

Virtual photons are not just part of the *dressing* of charged particles, where in standard quantum field theory they appear as aspects of the calculation, in PTQT they enable the transaction.

The table below summarises the key differences between real and virtual photons in standard quantum field theory:

Feature	Real photon	Virtual photon
On/off shell	On-shell	Off-shell
Appears where?	External legs	Internal lines
Detectable?	Yes	No
Energy–momentum relation	Satisfies $E^2 = p^2c^2$	Violates it
Role	Physical radiation	Mediates interactions mathematically
Lifetime	Arbitrary	Constrained by propagator structure
Physical status	Actual field excitation	Bookkeeping device in perturbation theory

Only real photons correspond to measurable radiation and virtual photons are indispensable in calculations but their physical existence is in the quantum substratum. Confusing the two leads to misconceptions like “vacuum is full of particles popping in and out of existence.” In

QFT, virtual particles are *terms in an integral*, not literal transient objects, but in PTQT they are essential for completing the transaction.

A virtual photon in the substratum enables the transaction between the Emitter States and the set of possible Absorber States. It does not select the specific absorber but does select a specific σ-algebra of possible absorbers. The selection of a specific absorber is a stochastic transition.

Ontological status of the virtual photon in the quantum substratum (PTQT):

- It is not a particle and not a spacetime object.
- It enables the transaction between the emitter and the selected absorber.
- It exists only in the quantum substratum, which is *not* spacetime and has no metric, no geometry, no trajectories.

Ontological status of the real photon in spacetime (PTQT):

- It is the actualised transfer of conserved quantities (energy–momentum).
- It is a spacetime connection between an emission event and an absorption event.
- Key properties:
  - Exists only after a transaction (NU-process).
  - Defines spacetime events — absorption events become actual spacetime nodes.
  - Carries energy–momentum — unlike the transaction photon.

– Has a spacetime trajectory (null geodesic) *only once* spacetime exists.

In summary photons in PTQT have the following features:

Feature	Photon in quantum substratum	Photon in spacetime
Ontological status	Possibility	Actualised energy transfer
Domain	Pre-spacetime substratum	Emergent spacetime
Trajectory	None	Null geodesic between events
Time structure	Time-symmetric process time (pre-spacetime)	Time-asymmetric spacetime time
Role	Enables potential interactions	Makes spacetime structure

6. Conclusion

6.1. Summary of the Argument

This book set out to reconstruct quantum mechanics on a probabilistic footing, using the σ-complex formalism of Chapter 1, and then to ask what this reconstruction implies for locality. Three distinct notions of locality have been distinguished:

- **Causal locality (microcausality)**: the requirement, inherited from special and general relativity, that operators associated with spacelike-separated regions commute (Section 2.1).
- **Bell locality**: the statistical independence assumption used in the derivation of the Bell/CHSH inequalities, whose violation by quantum mechanical predictions is confirmed by experiment (Chapter 3).
- **Haag locality**: the axiomatic requirement of local quantum field theory that observables be assigned to spacetime regions, together with microcausality and Haag duality (Section 4.6).

The central tension explored is that microcausality and Haag locality both presuppose a fixed spacetime background, while the violation of Bell locality shows that entangled systems are correlated in a way no local common cause can explain. Standard quantum mechanics accommodates this by remaining silent on the mechanism connecting preparation and measurement outcomes, deferring to the Born rule and to an unanalyzed notion of measurement.

The relational/probabilistic reconstruction developed here, together with the discussion of Possibilist Transactional Quantum Theory (PTQT) as an explicit ontological proposal, offers an alternative: entanglement and nonlocal correlation are located not in spacetime but in a pre-spacetime quantum substratum of physical possibilities, from which spacetime events emerge through the actualisation of transactions. On this view, Bell nonlocality is not a violation of causal locality, because the correlations do not require any propagating spacetime influence; and Haag locality, being a spacetime-dependent axiom, cannot be treated as fundamental if the substratum picture is correct.

6.2. What Has Not Been Resolved

This account is deliberately partial. Several difficulties have been identified rather than settled:

- The reconstruction in terms of σ-complexes and conditional states clarifies the mathematics of state reduction but does not by itself supply a dynamical mechanism for reduction; the treatment of the measurement problem shows what such a mechanism must do, not what it is.
- The relationship between PTQT and Haag's axiomatic local quantum physics is one of tension rather than reconciliation: the two frameworks answer different questions about locality, and no unified formalism has been presented that recovers Haag's local algebras as an emergent, approximate description of the substratum.
- The status of the quantum substratum itself — whether it should be understood as a further physical domain requiring its own dynamics, or as a heuristic device for speaking about unmeasured possibilities — has been described rather than derived from independent principles.

6.3. Open Questions

The discussion points toward several questions for further work:

1. Can the emergence of Haag-local spacetime algebras from a pre-spacetime substratum be given a precise mathematical treatment, rather than a conceptual one?
2. What experimental signatures, if any, would distinguish PTQT-type substratum theories from other interpretations (Bohmian mechanics, many-worlds, relational quantum mechanics) that also reproduce the Bell-violating statistics of standard quantum mechanics?
3. Does the probabilistic reconstruction extend cleanly to relativistic quantum field theory, where Haag's framework is best established, or does it require modification at that level?

These questions are not resolved here. The aim of this book has been to make the locality problem in quantum mechanics precise enough that such questions can be posed sharply, and to show that at least one substratum-based account offers a coherent, if incomplete, response to it.

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